Tutorial Lab 12.4 (Formatively Assessed):

Amplifier Use, Feedback, and Stability Analysis

Preliminary Matter: Setting up the Cadence Environment

Please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout before entering these commands into your terminal.

For this course, you will use the Cadence Virtuoso Design Environment (referred to simply as Cadence). Cadence is a Linux based software suite for designing full custom integrated circuits. In order to use Cadence you will first need to set-up the Cadence environment.

11. Open a terminal window and ensure that you are in your home directory using the following command:

```
cd ~
```

Note: 'cd' stands for change directory. The solitary tilde symbol (~) is a shortcut/alias for your home directory (/home/username).

12. Under your home directory, use the following commands to load the Cadence environment:

```
module use /eda/cadence/modules
module load course/EENGM6030
```

Note: The two 'module' commands load the required shell/Cadence environmental variables for this course (paths, names, aliases…) In particular, this includes a reference to the process design kit (PDK) used in this course, the GPDK180. You will need to do this every time you open a terminal, since the sourced environment only applies to the current terminal (until it's closed).

a. ***Note: This step only needs to be completed ONCE - If you already ran this script in lab 12.1, you don't need to run it again here.*** The first time you use Cadence, you will need to run the following command (script):

```
EENGM6030_setup
```

This will create a folder called AICD (short for Analogue Integrated Circuit Design) under your home directory. The AICD folder contains the files and directories needed to run Cadence. Running this setup script more than once will result in a warning message that the AICD folder already exists. It will then exit without making changes.

13. The AICD folder will be used for all your Cadence related activities for this course. **You should always make sure you are in your AICD folder before you launch Cadence.** To run cadence, use the following commands:

```
cd ~/AICD
virtuoso &
```

Note: The 'virtuoso &' command starts Cadence. The ampersand symbol (&) causes the Cadence process to run in the background freeing up the terminal for other commands the '&' symbol is a feature of the terminal's shell (bash) not Cadence.

1. Learning Objectives of this Lab:

From the previous lab, you should be able to:

- Obtain the transconductance g_m from both the large-signal characteristics/DC-sweep and the small-signal AC analysis. (Plot the dI_{DS}/dV_{GS} vs. V_{GS}, dV_{DS}/dI_{DS} vs. V_{DS}, and g_m vs. V_{GS}, r_{DS} vs. V_{DS}, and g_m/I_{DS} vs. V_{GS} using DC and/or AC analysis.)

- Understand the need to choose an appropriate DC operating point. Appreciate the trade-off between performance and power consumption in choosing an operating point. Understand the underlying relationships between DC operating point analysis, large-signal and small-signal analyses, DC and AC analyses.

- Use the Cadence in-built calculator to post-process simulation data. Import simulation data from the schematic into the expression buffer using the schematic selection tool. Manipulate the expression, apply functions from the function panel to the expression, and evaluate the buffer. Send the expression from the calculator to the ADE. Save ADE and Calculator state.

- Carry out a basic frequency-sweep simulation using an AC analysis. Understand the use of linear vs. logarithmic frequency sweep, and the dB power scale to plot gain. Use the calculator to convert and plot the linear magnitude response on the dB power scale à $dB20(A_V) = 20 \log A_V$, where $A_V = V_{OUT}/V_{IN}$.

In this lab, you will:

- Understand the importance of feedback in (analogue) electronic circuits, the potential for instability within such circuits, and the loop gain magnitude and loop gain phase as the criteria/metrics for stability.

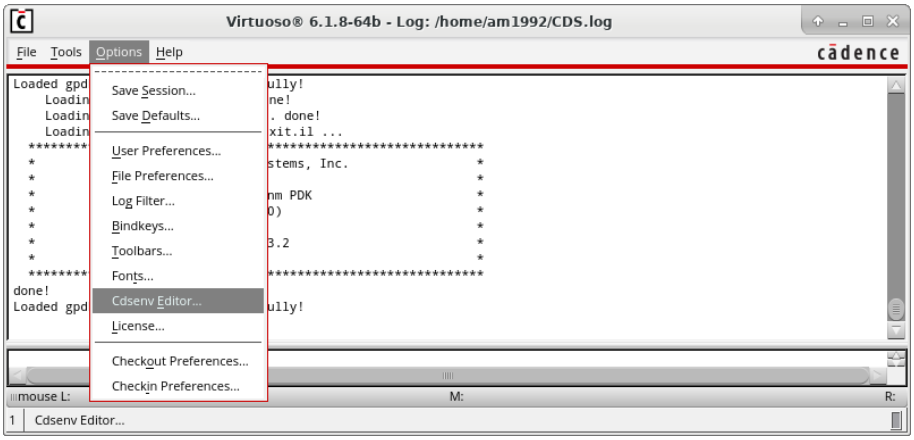
- Design and investigate a simple ideal feedback system. Investigate the SPICE/Spectre option for the analysis of stability in feedback circuits. Investigate the stability of a voltage reference circuit with modelled op-amp drive.

2. Cadence Documentation, Help, and Licensing

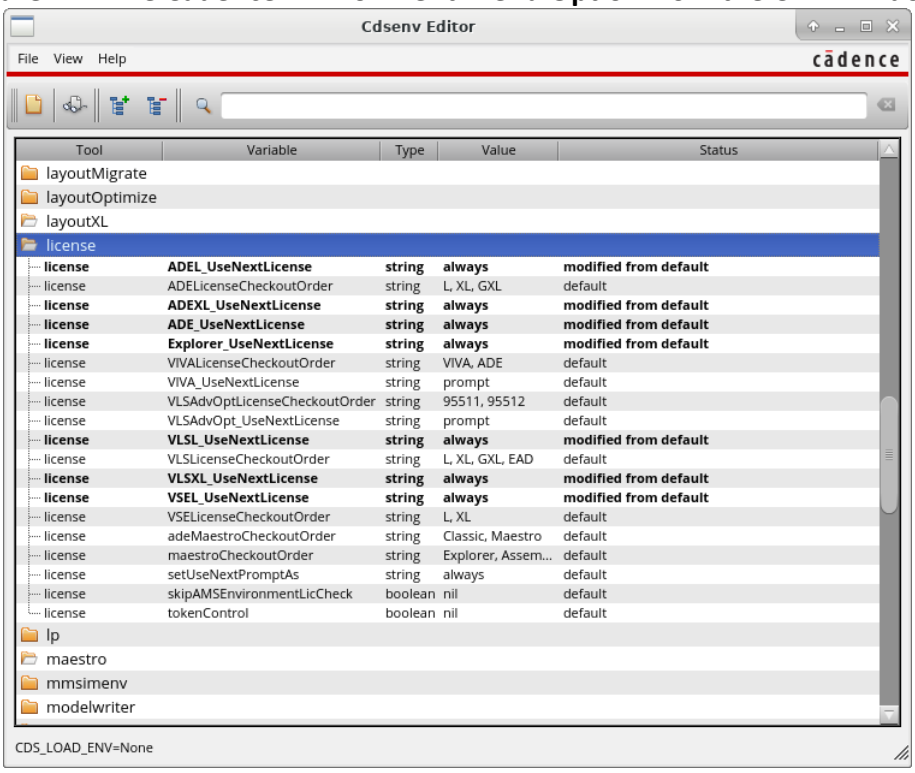
Local and online help is available from the help menus (**Menu: Help → User Guide** or by pressing **F1**), or by typing 'cdnshelp &' at the command prompt (assuming you have setup your terminal environment).

PLEASE NOTE: WHEN LAUNCHING A TOOL IN THE CADENCE DESIGN ENVIRONMENT YOU MAY SOMETIMES GET A WARNING ABOUT A LICENSE BEING UNAVAILABLE FOR THE TOOL, AND REQUESTING PERMISSION TO OBTAIN A HIGHER-TIERED LICENSE. YOU SHOULD SELECT 'ALWAYS' TO THESE MESSAGES.

If you accidently deny a license request, or you cannot access a license, you will need to modify the Cadence environment licence/permissions from the CIW. From the CIW menu go to **Menu: Options → Cdsenv Editor**. Once in the environment editor, go to the license folder, and make sure all licenses (in bold) are set to '**always'** – if any entries are listed as '**never**' you will need to manually change/retype them to '**always**'.



< Figure X.1. The Cadence Environment Menu Option from the CIW Window. >



< Figure X.2. The Cadence Environment Editor. Note, all licenses should be set to 'always'.>

3. Getting Started with Cadence:

For a refresher/introduction on how to start a Cadence session please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout.

3.1 Dealing with Lock files

PLEASE NOTE: IF A SCHEMATIC ISN'T PROPERLY CLOSED DOWN (FOR EXAMPLE, DUE TO A DROPPED REMOTE CONNECTION), THE SCHEMATIC MIGHT BECOME EDIT LOCKED. IF THIS HAPPENS YOU WILL NOT BE ABLE TO OPEN THE FILE FOR EDITING. YOU WILL ONLY BE ABLE TO OPEN THE FILE FOR READING UNTIL YOU REMOVE ANY/ALL LOCK FILES.

There are two ways you can unlock files:

(5) The first is to manually delete any edit lock files. To do this, go to your '~/AICD/Designs' folder and find the folder representing the schematic e.g., for lab1 this would be '~/AICD/Designs/lab121_vdivider/schematic' if this is what you called your schematic. Delete all the .cdslck lock files in this folder and you should be able to open your schematic for editing.

(6) You can also automatically release edit locks with the command:

```
clsAdminTool -are .
```

To use this command, you have to be in your AICD folder (the full-stop is shorthand for the current directory path). If you are not in your AICD folder, you have to specify the path (relative or absolute) to your AICD directory.

Since this is a Cadence command, you have to ensure that you have set up the terminal environment for Cadence (in particular, the modules must be loaded as in step 2, 'Preliminary Matter'). It is best to use the terminal window that you used to start Cadence, since the environment was already set up. Note that you will not be able enter any commands into your terminal if you did not start cadence with the '&' character. By example, to release any lock files after opening a new terminal, the full command set would be,

```
cd~
module use /eda/cadence/modules
module load course/EENGM6030_2021
clsAdminTool -are AICD
```

For more information on this command, use
`clsAdminTool -help`

4. Circuit Simulation Exercises: Part 1

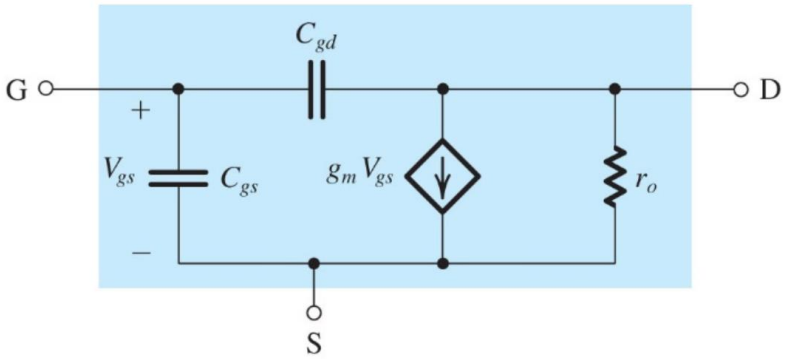
With guidance, you will:

- Understand the importance of feedback in electronic circuits for control, and the potential for instability within such circuits.
- Design and investigate a simple ideal feedback system.
- Investigate the SPICE/Spectre option for the analysis of stability in feedback circuits.

As we discussed in the previous lab, the AC analysis simulates an idealised linear model of the circuit under test – the so called small-signal model. The small-signal model of the transistor is built from ideal components – resistors, capacitors, and voltage-controlled current sources (VCCS) – representing the ideal as well as non-ideal behaviours of the circuit. The values of these ideal components are determined from a DC operating point analysis of the large-signal, non-linear circuit model.

At the end of the last lab, we carried out a frequency sweep AC analysis where we observed the frequency response of the basic resistor-loaded amplifier. We observed from the bode plot (magnitude/dB frequency response plot) that the circuit had at least one pole (at around 10^9 Hz) and one zero (at around 10^{11} Hz) caused by the parasitic reactances and resistances inside the model of the NMOS device. The most significant capacitances in the circuit are the gate-to-source c_{gs}, gate-to- drain c_{gd}, and drain-to-source c_{ds} capacitances. Due to the large-voltage variation across the gate-to-drain capacitance, it is this capacitance that dominates the frequency response and gives us the dominant pole responsible for the 3dB roll-off near 10^9 Hz. This same capacitance also gives us the zero near 10^{11} Hz (as the frequency increases the signal passes straight from input-to-output via this parasitic route through the transistor).

Note the small signal model from last week, with parasitic capacitances added.



< Figure 82. >

Fundamentally, the stability of a circuit is determined by the location of the poles and zeros of the circuit in the frequency domain. In this lab, we will see how the poles and zeros arise from the RC products in the circuit in both the forward and feedback paths of a simple

feedback system (which models an amplifier). We will explore how the poles and zeros affect the frequency response of the circuit – the input-output frequency response, as well as the frequency response of the feedback loop – and crucially what this means for the stability of the circuit. We will introduce loop gain magnitude and loop gain phase as the criteria and metric for amplifier stability.

4.1 Feedback and Stability

In the analogue IC designer's toolbox, feedback is the most useful tool. We use feedback to control our circuits – to set and maintain the bias/operating points, to compensate for non-linear and non-ideal behaviour (linearisation, compensation), to improve process-voltage-temperature (PVT) robustness, to reduce parameter sensitivity between devices and circuits, to increase or decrease input and output impedances, to compare/track/regulate/mirror currents and voltages, to shape the frequency response, and more.

But feedback is problematic. As soon as you introduce feedback into a circuit you introduce the potential for instability. When utilising feedback, we almost always use **negative feedback** – where the feedback signal is subtracted from the input signal and the difference (the error signal) is used to drive the system. When a system feeds the output back to the input such that it adds to the input, you have **positive feedback**. The output, in this case, will grow unbounded until it saturates (reaches some system limit). Consider the **Barkhausen criteria** for instability (aka Barkhausen stability criterion, Barkhausen criterion for sustained oscillation). A system that satisfies the Barkhausen criteria will oscillate – in fact, the Barkhausen criteria is the defining criteria in oscillator design. Beyond the Barkhausen criterion the system is unstable – it will not stabilise or settle at an operating point.

We must therefore ensure when using feedback in a system, that it remains stable over a very wide frequency range. If the circuit is unstable at any point in the frequency domain, it is considered unstable overall. Even where instability occurs outside of a system's intended frequency range of operation, disturbances, noise, or other unintended/unforeseen inputs outside of the specification may drive the system into instability.

To determine the stability of a circuit through simulation requires, a **stability analysis**, which is separate and distinct from the AC and DC analyses. You cannot determine whether a circuit is stable through AC or DC analysis alone. The DC analysis does not simulate time-domain or frequency-domain related behaviour such as oscillations. The AC analysis considers only the frequency domain steady-state behaviour. It is possible to spot instability in a transient analysis in the form of oscillations and exponentially increasing responses, but this does not help in characterising where the instability lies. The only meaningful way to determine if a circuit is stable is through a stability analysis. In Cadence, this is the '**stb**' analysis type. Note that the stability analysis is a small-signal analysis – it simulates an idealised linear circuit for a given DC operating/bias point. Using the stb analysis, determination of stability is made from the observation of the magnitude and phase response of the feedback loop. Specifically, we look for the point at which the feedback transitions from negative to positive.

For a fuller explanation of feedback theory refer section 5.1 of the background material and reference [3] – Appendix E (Sedra/Smith).

Operational Amplifier (Op-Amp) Circuits

Op-amp circuits are the principal example of a feedback system. The general structure of the op-amp is a **differential input stage** followed by a **high gain stage**. (Usually, the gain stage has a high output-impedance, and may be followed by a low-impedance unity-gain buffer stage.) The input stage has an inverting and non-inverting terminal – feedback is applied by taking the output (or some fraction/function of the output) and feeding it back to the inverting input. The feedback is then a comparison between the feedback signal and the input signal. Mathematically, we have a positively sensed input and a negatively sensed feedback signal. Note how this correlates to the basic model of the feedback system.

The op-amp demonstrates a core design philosophy in analogue IC design – in many cases we prefer not to create amplifiers with absolute parameter values, where due to PVT, there can be large tolerances in the values. Instead, we use a high-gain amplifier with feedback. The operation (linear gain, filtering, differentiation, integration, summing, etc.) is then set by the arrangement/configuration of external passive devices (external to the amplifier not to the IC) – namely resistors and capacitors. In this case, precision of the circuit is often determined by the ratios (and therefore the matching) of the external passive devices than by the absolute values of the passive devices. The matching of two or more devices in an IC can be done with far more accuracy compared to the absolute values. For example, consider the closed-loop gain of the inverting and non-inverting operational amplifiers. With a sufficiently high op-amp gain,

$$A_{V(-)} = -\frac{R_F}{R_{IN}} \qquad\qquad A_{V(+)} = 1 + \frac{R_F}{R_{IN}}$$

where R_F and R_{IN} are the feedback and input resistors. It is the ratio of these two devices that set the gain and not their absolute values. Moreover, when subjected to variations in process, voltage, and/or temperature as long as the ratio is maintained (using proper layout techniques), the gain accuracy is maintained. The combination of high gain and feedback can overcome many non-idealities in the circuit.

Dependent/Controlled Sources

There are four general amplifier topologies – defined by their input-output quantities: voltage amplifiers (voltage-in, voltage-out, gain=V_{OUT}/V_{IN}), transconductance amplifiers (voltage-in, current-out, gain=I_{OUT}/V_{IN}), transimpedance amplifiers (current-in, voltage-out, gain=I_{OUT}/V_{IN}) and current amplifiers (current-in, current-out, gain=I_{OUT}/I_{IN}).

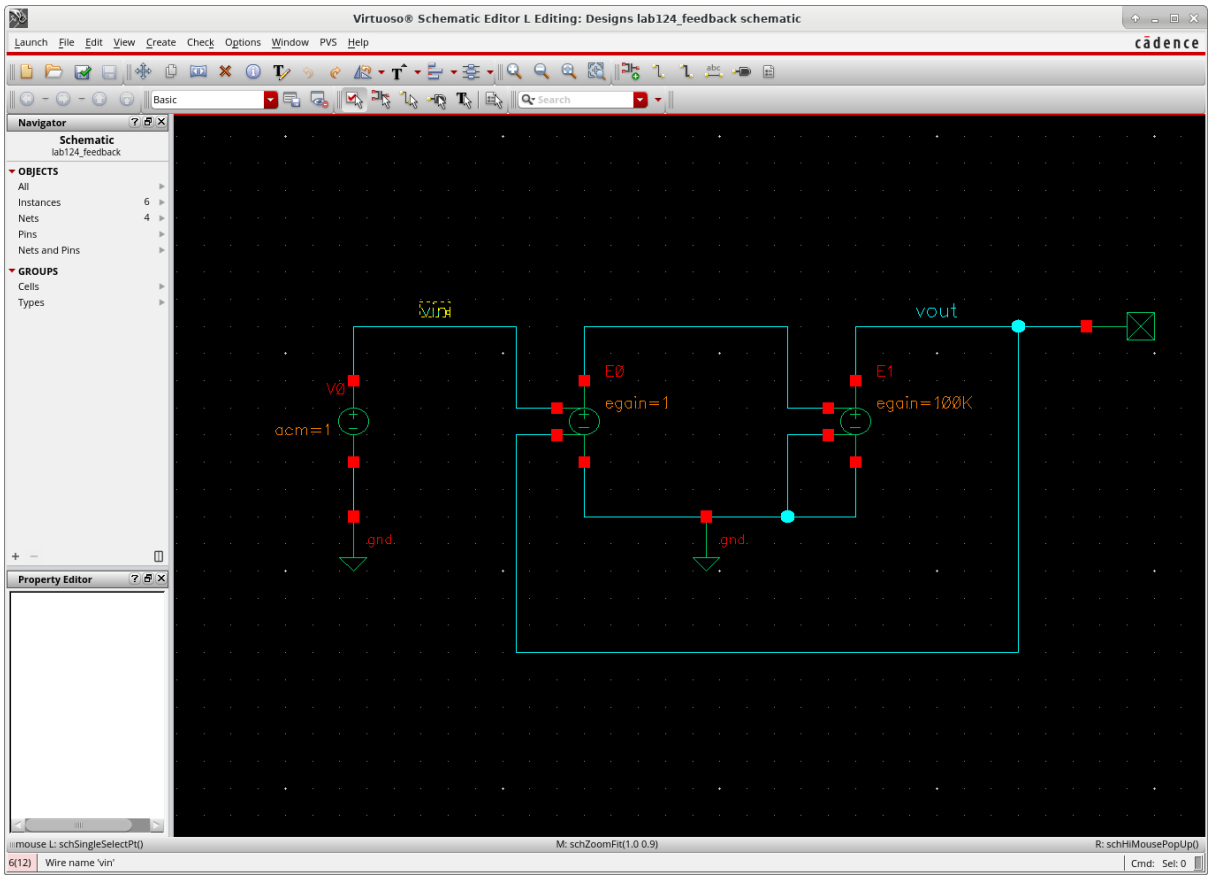
Each of the four amplifier types can be modelled by a **controlled source** from the analogLib library. Each of these controlled sources are ideal and have a linear scalar input-to-output relationship: voltage-controlled voltage-source (VCVS, voltage-gain=V_{OUT}/V_{IN}), voltage-controlled current-source (VCCS, transconductance=I_{OUT}/V_{IN}), current-controlled voltage source (CCVS, transresistance=V_{OUT}/I_{IN}), and current-controlled current source (CCCS, current-gain=I_{OUT}/I_{IN}). In this lab, we will use the VCVS to design and explore a simple feedback system. Note that we can use the VCVS to represent the Thevenin equivalent circuit of an amplifier, while the VCCS can be used to represent the Norton equivalent circuit of an amplifier.

4.2 A Simple Ideal Feedback System

(4.2.1) Create a new schematic. From the Library Manager, **Menu: File à New à Cell View**. Ensure you are in your Designs library. Use the New File form as you have done previously, giving your new schematic a meaningful name such as 'lab124_feedback'.

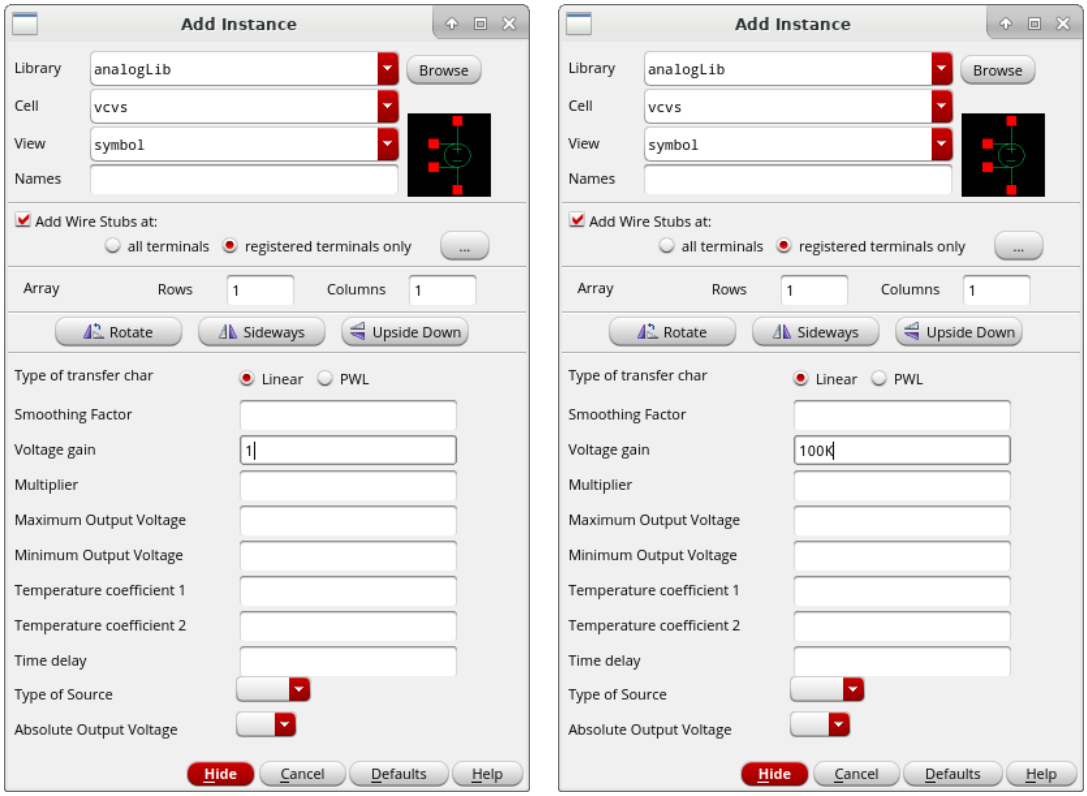
(4.2.2) In the schematic editor window, create the unity feedback system as given in Figure 2 as described below.

(4.2.3) From the analogLib library, instantiate two voltage-controlled voltage sources (**vcvs),** a DC voltage source (**vdc**), and a ground reference (**gnd**). The first VCVS in the feedback system will be used as the differential input stage with unity gain, set the voltage gain parameter to 1 (**Voltage gain** = 1). The second VCVS will be used as the high gain stage, set the voltage gain to 100K (**Voltage gain** = 100K). We will use the vdc component to provide the input ac stimulus, set the ac magnitude to 1 (**AC magnitude** = 1). Leave all other parameters/attributes empty. See Figures 2, 3 and 4.

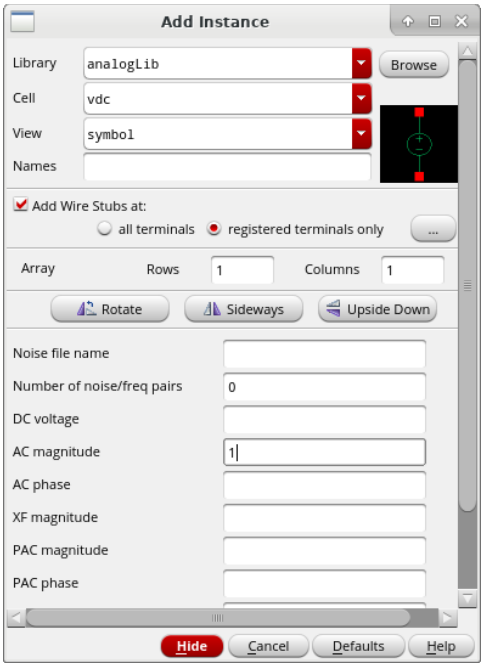


< Figure 83. Simple Feedback System. Ideal op-amp model using two VCVS components from the analogLib library. Unity gain configuration. >

(4.2.4) Wire the components as in Figure 2 using the **Wire (narrow)** command. Connect the input voltage source to the positive terminal of the differential input stage (+vcvs) and feedback the output to the negative terminal of the input stage (-vcvs). Name the output net *vout* and the input net *vin* using the **Create Wire Name** command. Use the **noConn** connector from the **basic** library to avoid floating-net warnings. Keep the schematic neat.



< Figure 84. Add Instance forms. VCVS for the input stage and gain stage, respectively.>

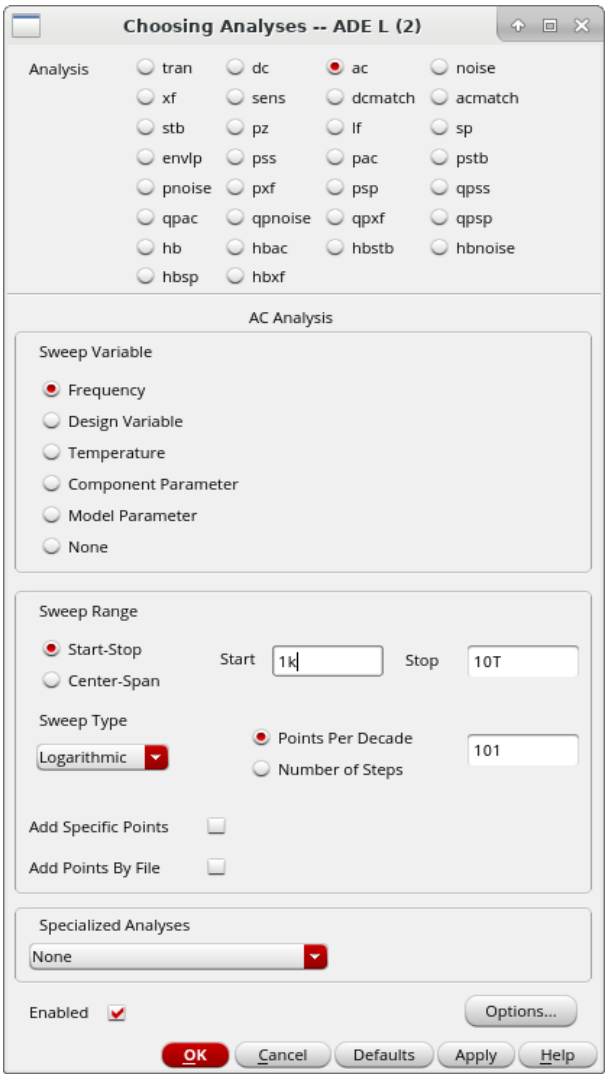


< Figure 85. Add Instance forms VDC for the AC input stimulus. >

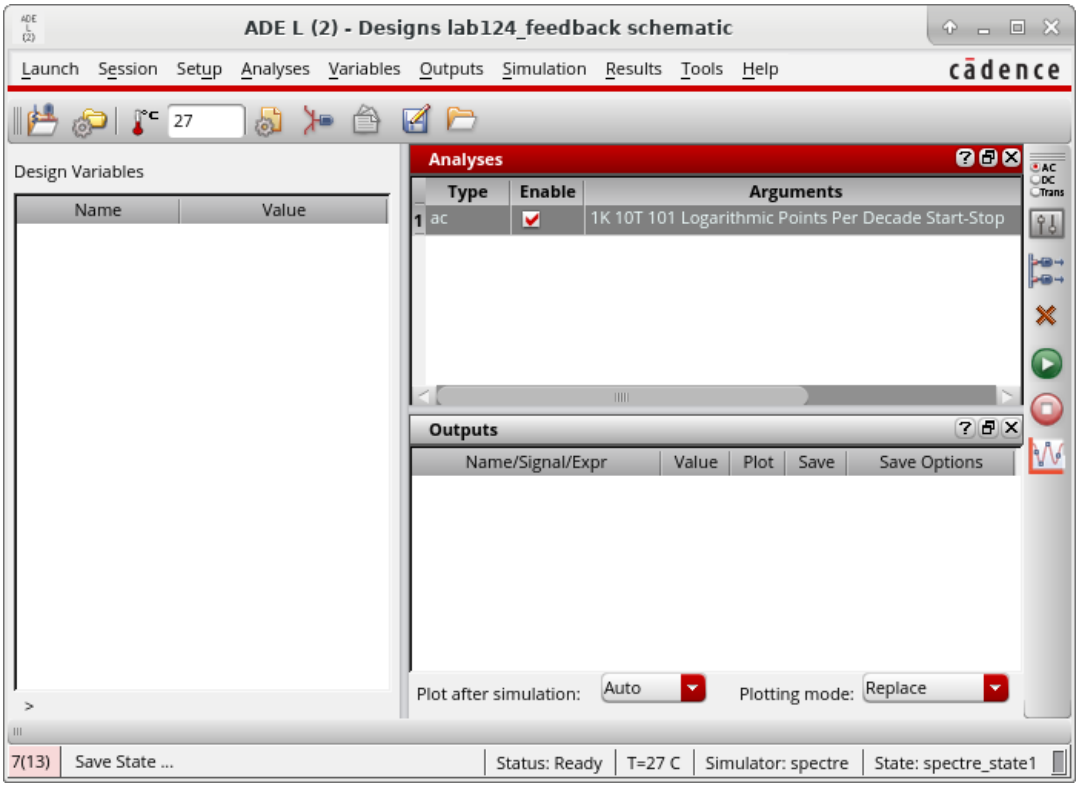
(4.2.5) Check and save your schematic using the **Check and Save** command. Ensure that it checks and saves without warnings or errors. Use the output area of the CIW to diagnose any problems with your schematic and/or simulations if there are any warning or error messages when checking, saving, or simulating your work.

(4.2.6) From the ADE window, setup an AC frequency sweep analysis (**Menu: Analyses à Choose**). In this simulation we will sweep the frequency from 1KHz to 10THz. Because of the large frequency range, we will use a logarithmic sweep of the frequency – with 101 plot points per decade of frequency. At this stage, we will not select any outputs for plotting.

In the choosing analyses form, select the '**ac**' option button. Select the 'Frequency' option as the sweep variable. Choose the 'Start-Stop' option button with Start = 1K and Stop = 10T for the sweep range. Select 'Logarithmic' from the drop-down menu as the sweep type. Choose the 'Points Per Decade' option button with 101 plot points per decade. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. OK the form. The **ac** analysis should appear in the 'Analyses' list of the ADE window. See Figures 5 and 6.



< Figure 86. The Choosing Analyses Form for an AC frequency sweep. >



< Figure 87. The ADE with the only AC Analyses set up. >

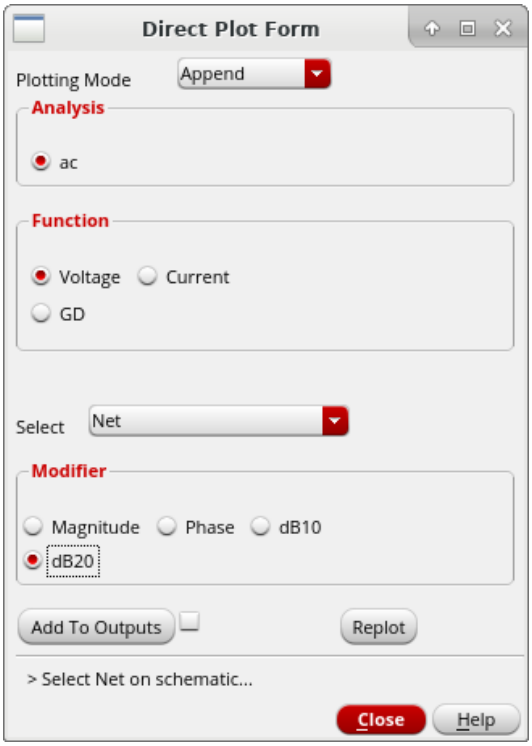
(4.2.7) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error. Only the text output window should open automatically since we didn't select an output for plotting.

The Direct Plot Form

(4.2.8) Note: Until now, every time we have run a simulation we have set up the ADE, *before the simulation is run*, using the 'To Be Plotted' command (Menu: Outputs → To Be Plotted → Select On Schematic). This would plot one or more outputs in the waveform window upon completion of the simulation. A more general approach is to plot the outputs manually, *after a simulation has run*, using the **Direct Plot** command of the ADE. Using this option, you can quickly probe the circuit – selecting and plotting one or more waveforms at different nets and nodes one by one – without having to set them up in the ADE beforehand. You can also plot common expressions quickly without having to use the calculator.

(4.2.9) From the ADE window, select **Menu: Results → Direct Plot → Main Form** to invoke the Direct Plot Form (this may also invoke an empty waveform window). In the 'Analysis' section of the Direct Plot Form there will be an option button for every analysis type setup and run in the ADE. In this case, since we have only setup and run an AC analysis, there is only an '**ac**' option present. Associated with each analysis type (e.g., *trans, ac, dc, stb…*), will be a number of plotting options; for the ac analysis we have the function options (e.g., voltage, current, group delay) and the modifier options (e.g., magnitude, phase, dB10, or dB20). The options available can be very different for the different analysis types.

At the very top of the form is the potting mode drop-down menu, with the usual options – Replace, Append, New Sub-Window, and New Window. Leave it on the 'Append' option.



< Figure 88. The Direct Plot Form for AC Analysis. >

(4.2.10) From the Direct Plot Form, plot the output voltage magnitude and phase. Make sure the **ac** option button is selected in the analysis section. Select the **voltage** option button from the function section. In succession,

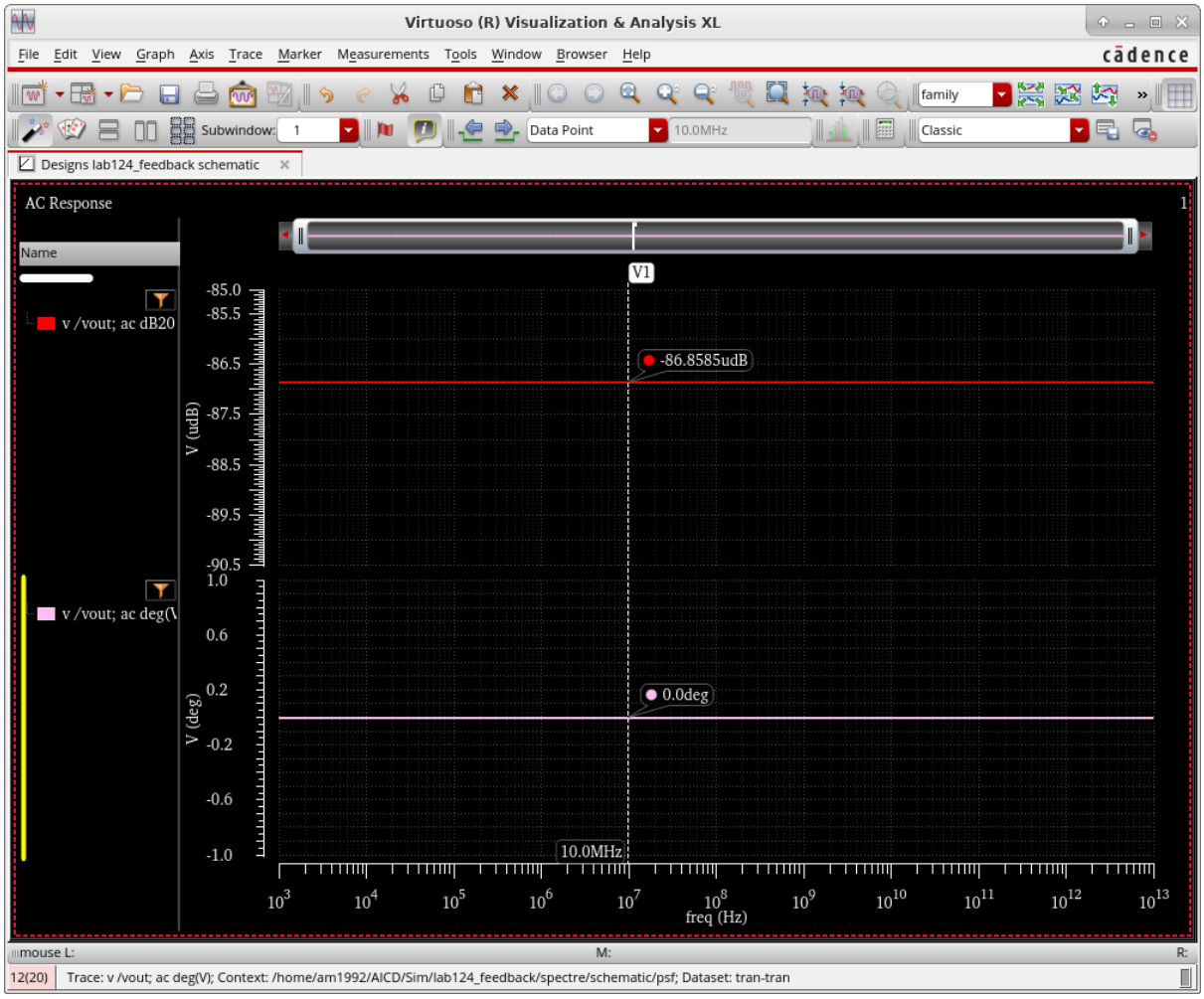
(1) Select the **dB20** option in the modifier section of the direct plot form. Then, in the schematic window, click on the output net *vout*. The output voltage magnitude plot vs. frequency should appear in the waveform window.

(2) Select the **phase** option in the modifier section of the direct plot form. Then, in the schematic window click again on the output net *vout*. The output voltage phase plot vs frequency. should be added to the waveform window (assuming the form's potting mode is set to append).

You should now have both a plot of amplitude (dB20) vs. frequency and a plot of phase (degrees) vs. frequency plotted in the waveform window. See Figure 8.

(4.2.11) From the schematic editor window, **press the 'ESC' key to exit the plotting mode of the Direct Plot Form**. Or simply close down the Direct Plot Form.

(4.2.12) Split the traces for clarity. Right-click on the empty area of the graph window and select **Split Current Strip à trace** from the context menu.



< Figure 89. Plot of the Amplitude (dB20) vs. Frequency, and Phase (degrees) vs. frequency.
1KHz to 10 THz >

(4.2.13) **Discussion**. Referring to Figure 8. The feedback system we are simulating, models a basic high gain op-amp connected in a unity-gain feedback closed-loop configuration. In this case, the closed-loop gain should be very close to one (linear scale) or 0dB (decibel scale). Recall, that the closed-loop gain *G* is given by,

$$G = \frac{A_V}{A_V + 1}$$

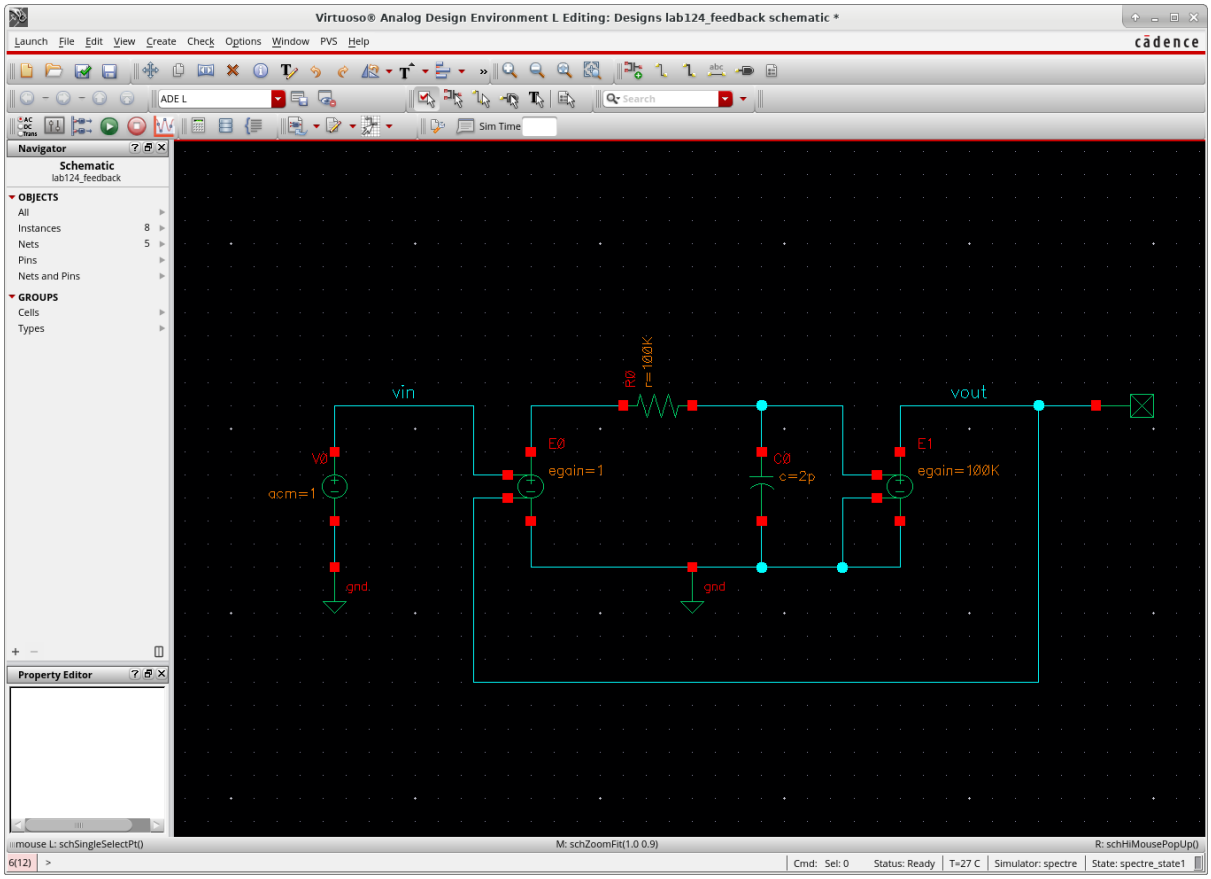
where A_V is the open loop gain of the system (100K). From the definition of dB20,

$$dB20(G) = 20\log G = 20\log\left(\frac{100K}{100K + 1}\right) = -86.86\ \mu dB$$

We see from Figure 8 (top), that the gain is indeed very close to unity (-87udB). It is not quite unity, since this would require an infinite open loop gain. We also note that since there are no reactive components in the model, the output response – magnitude and phase – has no frequency dependence i.e., zero degrees phase shift at the output across all frequencies.

4.3 Adding a Pole to the Forward Path

(4.3.1) Add a resistor and capacitor to the forward path of the simple feedback system as shown in Figure 9. From the analogLib, instantiate a resistor (**res**) and capacitor (**cap**). Set the resistor to 100KΩ (**Resistance** = 100K) and set the capacitor to 2pF (**Capacitance** = 2p).



< Figure 90. Simple Feedback System. Including a pole – caused by the RC low pass filter – in the forward path. >

(4.3.2) Check and save your schematic using the **Check and Save** command.

(4.3.3) Re-simulate the schematic. Again, use the Direct Plot form to plot both the magnitude (dB20) vs. frequency and phase (degrees) vs. frequency. See Figure 10.

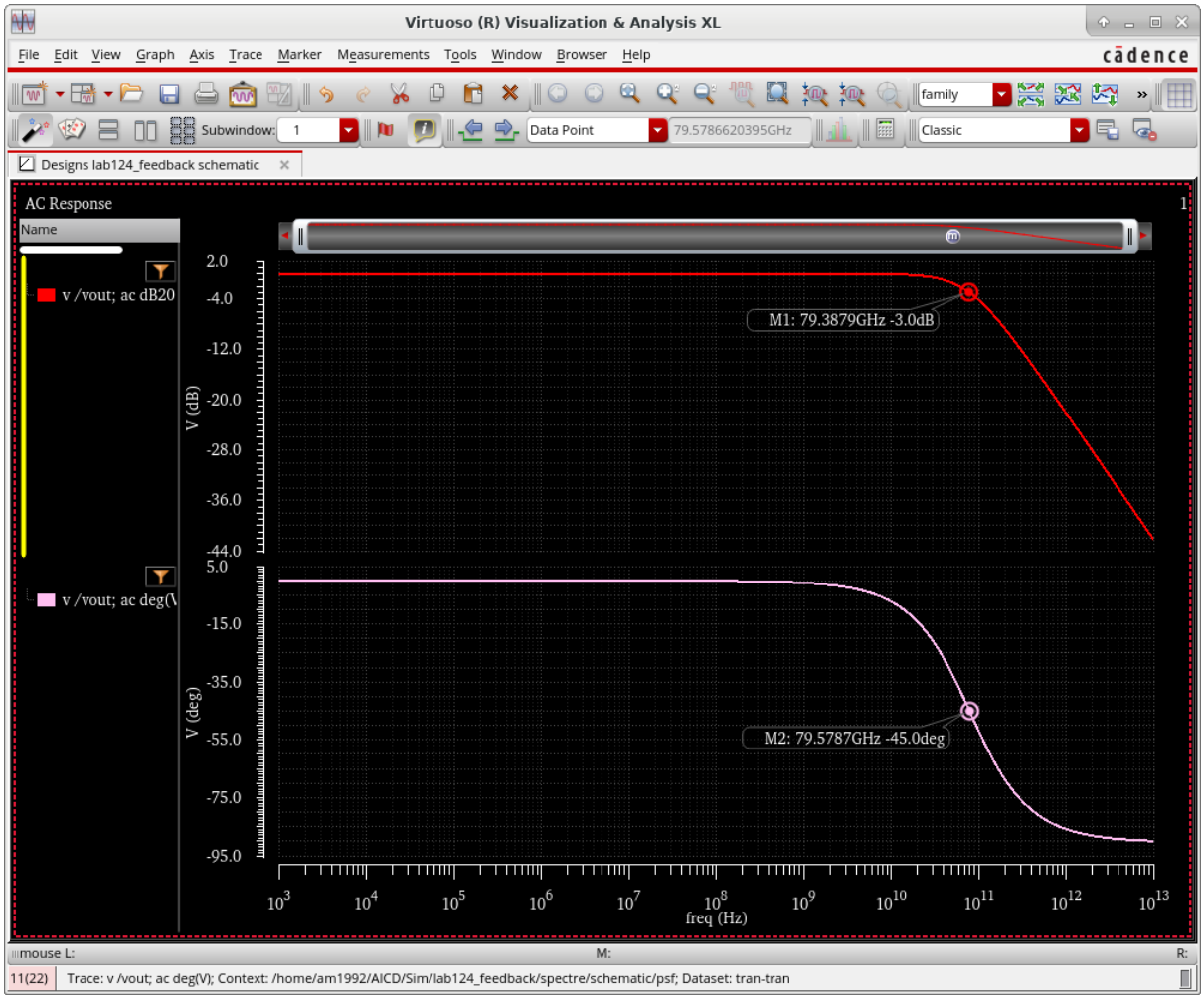
(4.3.4) Do not close down the waveform window.

(4.3.5) **Discussion**. Referring to Figure 10. We have introduced a low-pass filter into the forward path of the feedback system. The resistor/capacitor combination models a pole formed by the high output impedance of the first stage of a FET amplifier and the gate capacitance at the input of the following stage. Note that the combination of VCVS and series resistor represents the Thevenin equivalent circuit of the first stage's output – we could also do this with a VCCS and a parallel resistor (the Norton equivalent).

With the pole introduced in the forward path, caused by the RC product, we now see a frequency dependence in the output magnitude and phase response. In this case, the single pole leads to a corner-frequency located at:

$$f_{-3dB} = A_V \times \frac{1}{2\pi RC}$$

where R = 100KΩ, C = 2pF, and A_V = 100K, f_{-3dB} ≈ 79.58GHz. A marker is placed at the -3dB point on the magnitude plot and the -45° point on the phase point. See Figure 10.



< Figure 91. Bode plots for the simple feedback system with pole in the forward path. Plot of the amplitude (dB20) vs. Frequency (top), and Phase (degrees) vs. frequency (bottom). 1KHz to 10THz. >

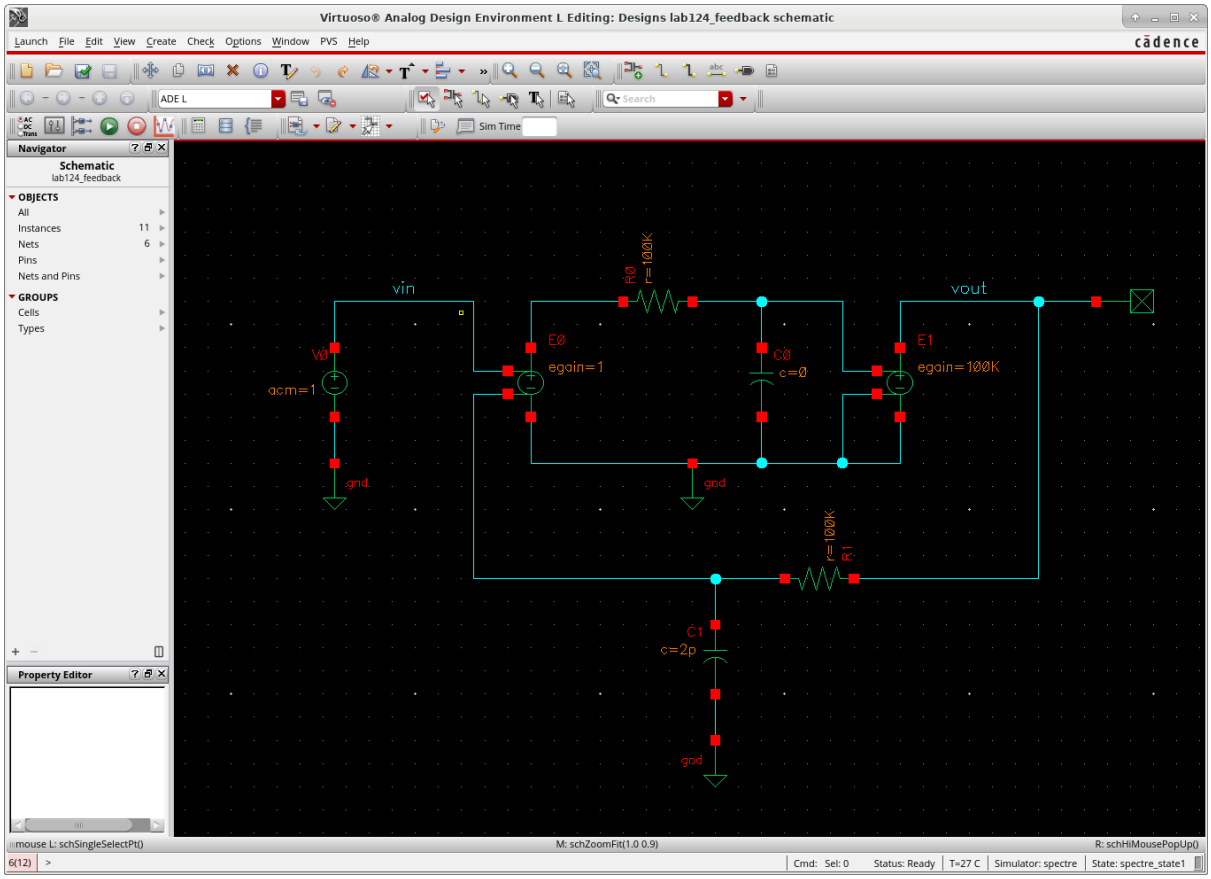
4.4 Adding a Pole to the Feedback Path

(4.4.1) Remove the capacitor in the forward path. Set the capacitor to 0 (**Capacitance** = 0). This effectively makes the capacitor look like an open circuit. The 100K resistor in the forward path has no effect since the input impedance of the following VCVS is infinite.

(4.4.2) Now, add a resistor and capacitor (low-pass filter) to the feedback path of the feedback system as shown in Figure 11. From the analogLib instantiate a resistor (**res**) and capacitor (**cap**). Set resistor to 100KΩ (**Resistance** = 100K), set capacitor to 2pF (**Capacitance** = 2p).

(4.4.3) Check and save your schematic using the **Check and Save** command.

(4.4.4) Re-simulate the schematic. Again, use the Direct Plot form to plot both the magnitude (dB20) vs. frequency and phase (degrees) vs. frequency. Make sure the plotting Mode of the Direct Plot Form is set to 'Append'.



< Figure 92. Simple Feedback System. Including a pole caused by the RC low pass filter – in the feedback path. Note the forward pole is removed since C = 0. The 100K resistor in the forward path has no effect since the input impedance of the following VCVS is infinite.>

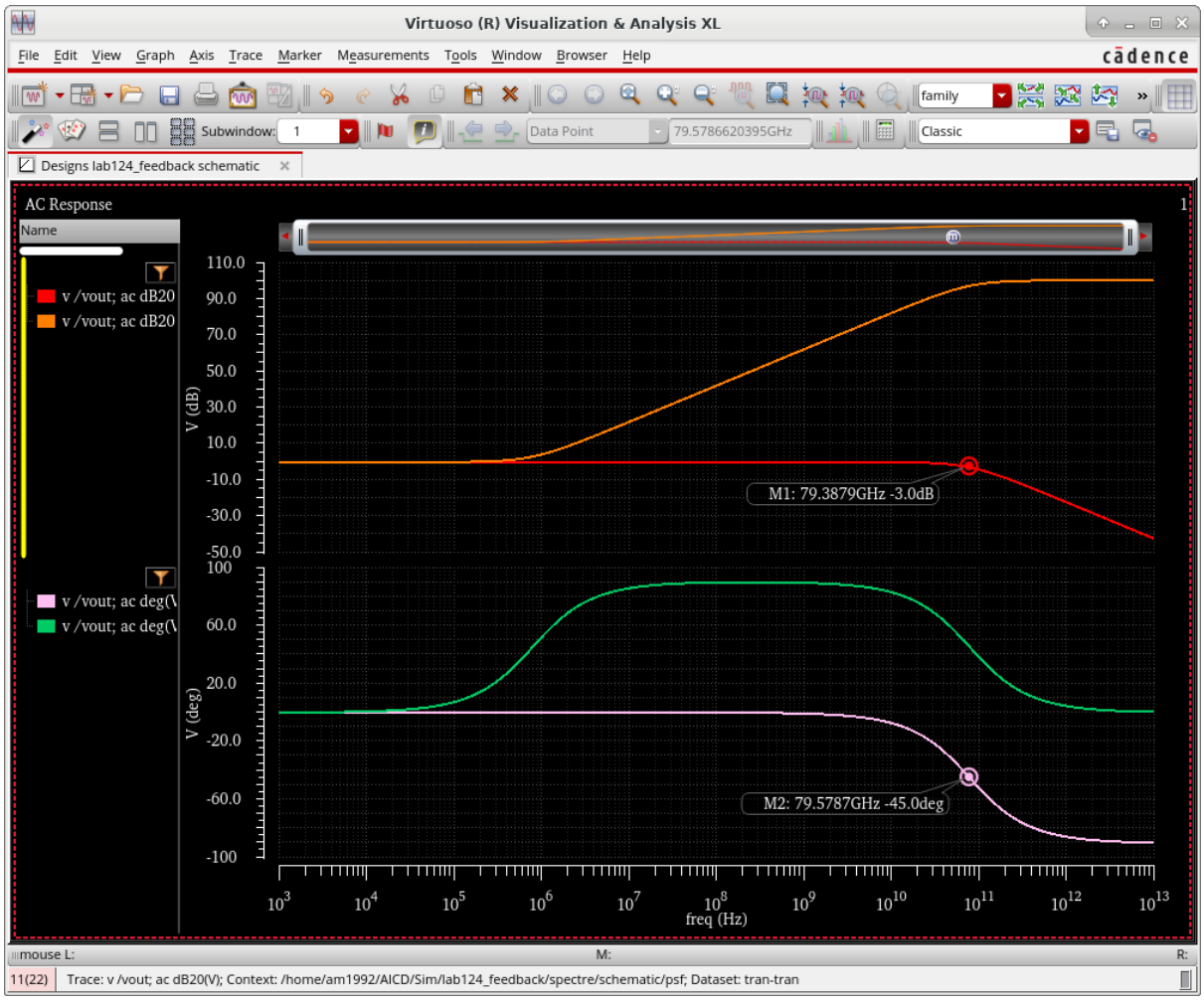
(4.4.5) **Discussion**. Referring to Figure 12 – note that the plot of the response with just the feedback pole has been appended to the plot of the response with just the forward pole. Here, we can see that the pole in the feedback path leads to a zero in the forward path located at,

$$f_1 = \frac{1}{2\pi RC}$$

This causes the output to behave like a high-pass filter. At low frequencies, below the zero frequency, the feedback system behaves like a unity-gain amplifier, with 100% of the output being fed back to the input. As the frequency increases, the pole (low-pass filter) in the feedback path begins to reduce the amount of signal being fed back to the input. As the level of feedback is reduced, the feedback system behaves more like an open-loop system. The apparent pole located at,

$$f_2 = A_V \times \frac{1}{2\pi RC}$$

occurs due to the finite open-loop gain of the amplifier. Above this frequency, the feedback pole no longer passes the feedback signal - the system effectively becomes open-loop with a gain of A_V=100K (=100 dB).

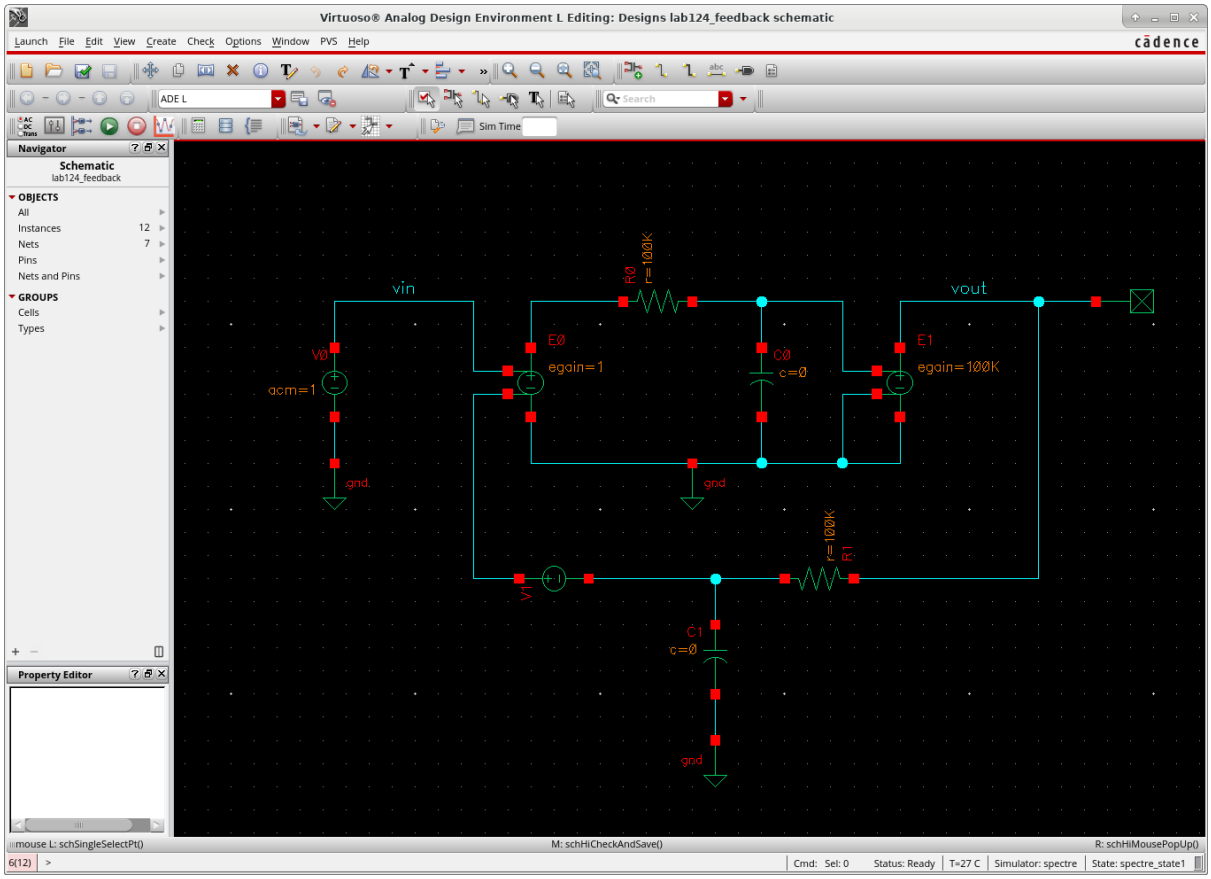


< Figure 93. Bode plots for the simple feedback system with pole in the feedback path. Plot of the magnitude (dB20) vs. Frequency (top), and Phase (degrees) vs. frequency (bottom). 1KHz to 10THz. Appended to previous plot of only forward path pole>

4.5 Stability Analysis: Loop Gain, Loop Phase, Phase Margin, and Gain Margin

(4.5.1) Here, we will explore the stability of the feedback system. Remove the capacitors in both the forward and feedback paths by setting both the capacitors to 0 (**Capacitance** = 0).

(4.5.2) Insert a null **vdc** voltage source in the feedback path after the capacitor Null means 0V, it is a short circuit element. The AC and DC magnitude values of this component should be set to zero or left empty. This vdc does not in any way change the functionality of the circuit or its output response. It is used only as a loop identifier/probe by the stability analysis tool.



< Figure 94. Simple Feedback System with all poles zeroed (C0=C1=0). A null vdc is placed in the feedback path after the capacitor (AC/DC Voltage = 0). It is used by the stability analysis to identify the feedback loop. >

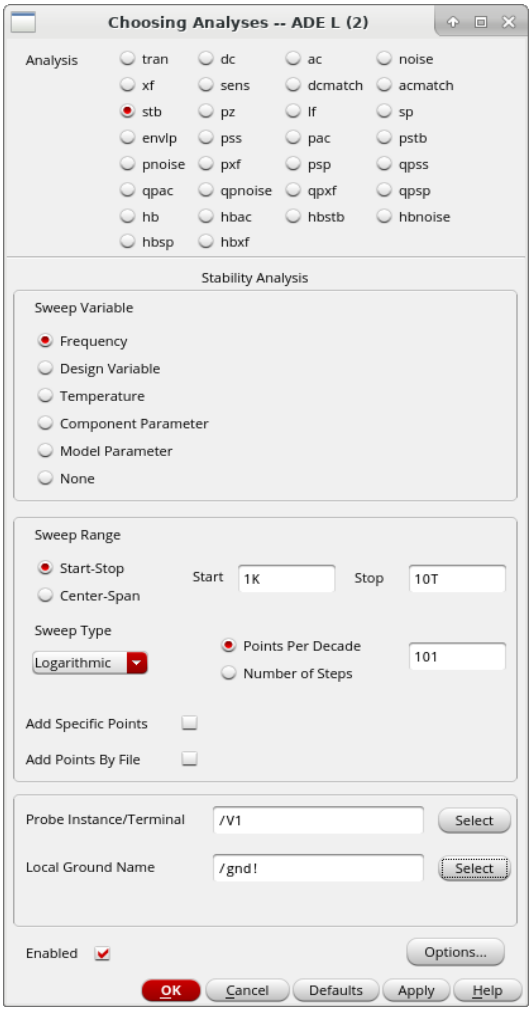
(4.5.3) We want to determine the stability of the feedback system. We do this by examining how the loop gain and phase of the circuit change with frequency. Specifically, we are looking for when the loop gain undergoes a 180° phase-shift, at which point if the gain magnitude is greater than unity (0 dB), the applied negative feedback becomes positive. As we discussed in the opening section (4.2), when the feedback becomes positive, the output will grow unbounded until it saturates. Note that the loop gain is the total gain around the feedback loop – it is the product of the forward gain ***A(ω)*** and the feedback factor ***B(ω)***. Refer to Figure 31, section 7.1.1. Recall, from the Barkhausen stability criteria that, stability is determined by the product ***AB*** being below unity (0dB) before the loop gain phase-shift reaches 180°.

(4.5.4) From the ADE window, setup a stability analysis (**Menu: Analyses à Choose**). In this simulation we will again sweep the frequency from 1KHz to 10THz using a logarithmic sweep of the frequency – with 101 plot points per decade of frequency. At this stage, we will not select any outputs for plotting (we will use the Direct Plot Form later).

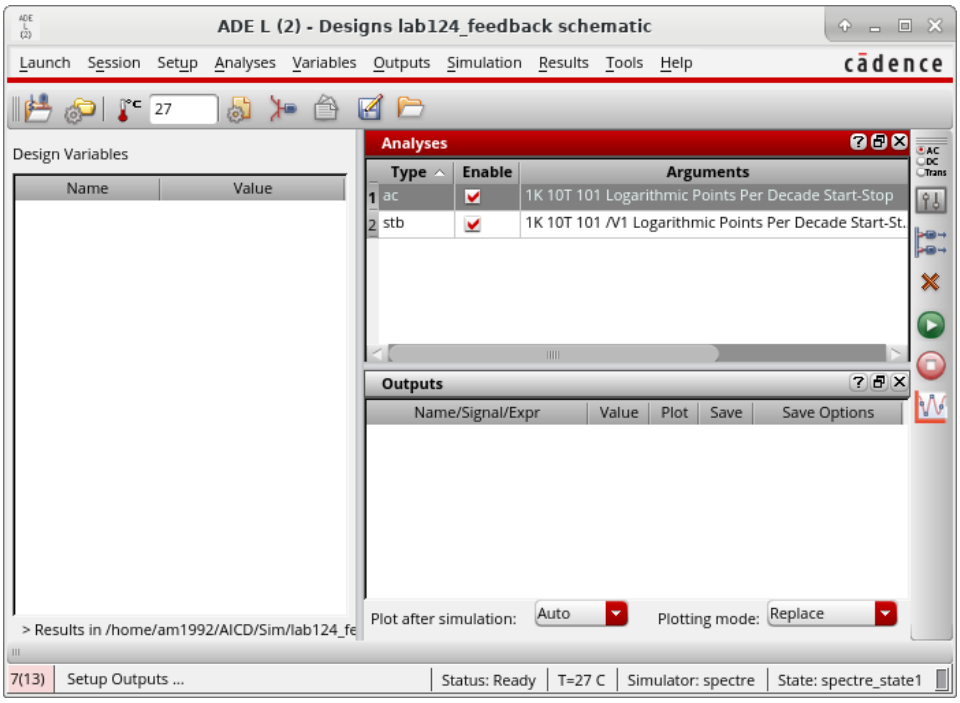
In the choosing analyses form, select the '**stb**' option button. Select the 'Frequency' option as the sweep variable. Choose the 'Start-Stop' option button with Start = 1K and Stop = 10T for the sweep range. Select 'Logarithmic' from the drop-down menu as the sweep type. Choose the 'Points Per Decade' option button with 101 plot points per decade.

At the bottom of the form, there are two fields: 'Probe Instance/Terminal' and 'Local Ground Name'. The probe instance/terminal is used to identify the feedback loop – we need to point this to the null vdc component we inserted into the feedback path. Press the select button next to the 'Probe Instance/Terminal' form field. The schematic should come into focus. Click on the null vdc component. The name of the null vdc part (/V1) should now appear in the form field. We also need to do the same with the local ground name. Press the select button next to the 'Local Ground Name' form field. The schematic should come into focus. Click on any net connected to a ground (gnd) reference. The name of the ground net (/gnd!) should now appear in the form field.

Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. OK the form. The **stb** analysis should appear in the 'Analyses' list of the ADE window next to the **ac** analysis already set up. See Figures 14 and 15.



< Figure 95. The Choosing Analyses Form for a Stability Analysis. >



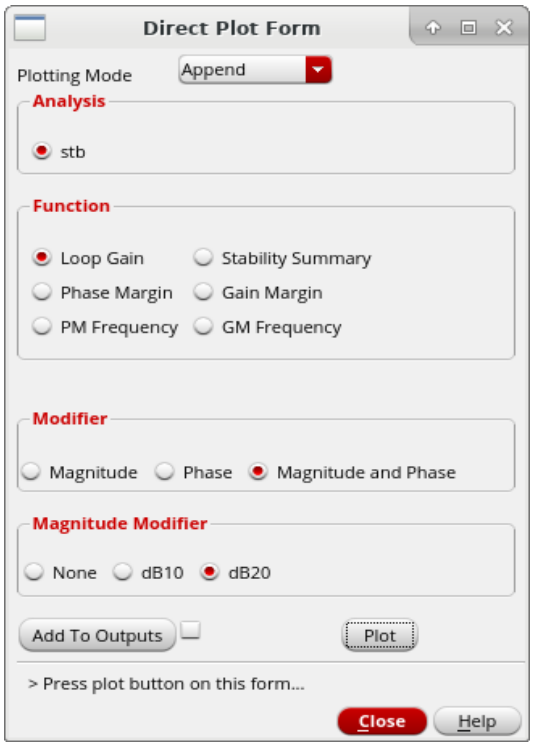
< Figure 96. The ADE with the AC and STB Analyses set up. >

(4.5.5) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error. Only the text output window should open automatically since we didn't select an output for plotting.

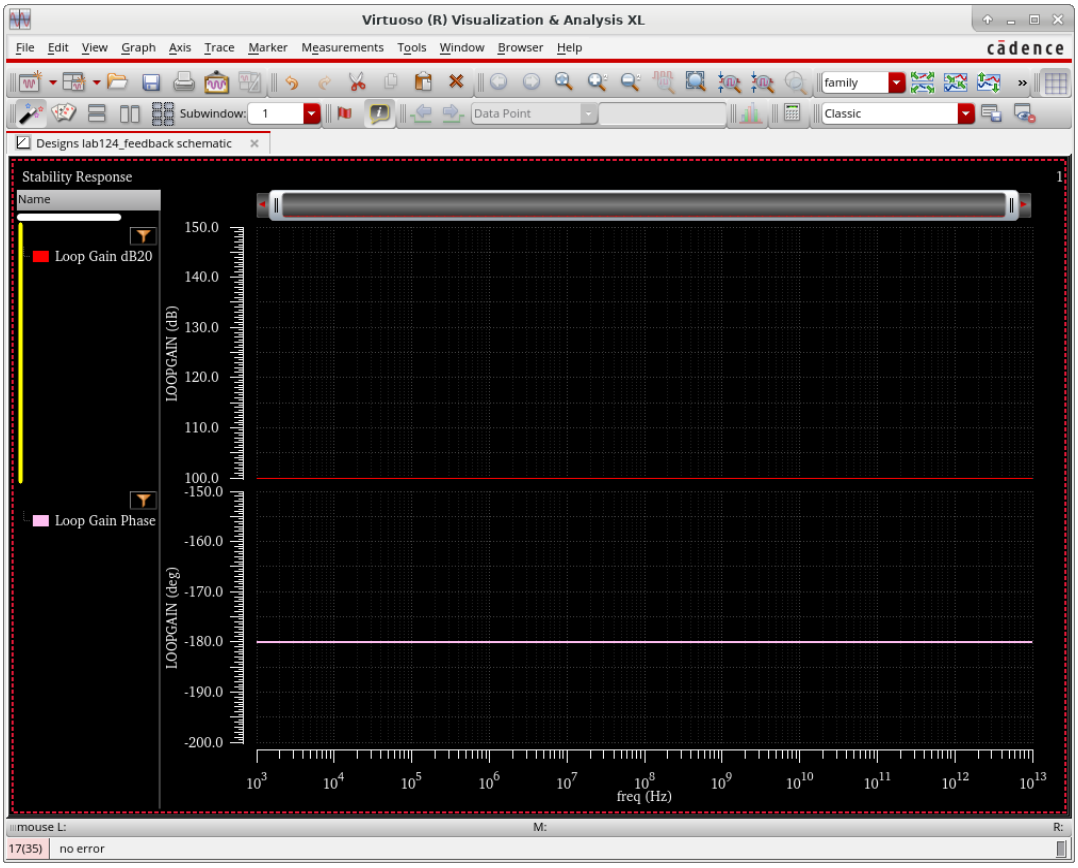
(4.5.6) From the ADE window, open the Direct Plot form (**Menu: Results → Direct Plot → Main Form**). Note that the **stb** analysis option button has been added alongside the existing **ac** option button to the 'Analysis' section of the Direct Plot form. Switch between the ac and stb options to see how the plotting options/expressions change depending on analysis type. For the stb analysis type, there are several plotting options for determining stability. At this stage we only want to plot the loop gain – magnitude and phase – versus frequency.

Make sure the **stb** option button is selected in the Analysis section of the Direct Plot Form. Select the **Loop Gain** option button from the function section. Select the **Magnitude and Phase** option in the modifier section of the direct plot form. In the magnitude modifier section, select the **dB20** option. Click the 'Plot' button near the bottom right corner of the form. You should now have both a plot of loop gain magnitude (dB20) vs. frequency and a plot of loop gain phase (degrees) vs. frequency plotted in the waveform window. See Figures 16 and 17.

(4.5.7) Split the traces for clarity. Right-click on the empty area of the graph window and select **Split Current Strip à trace** from the context menu.



< Figure 97. The Direct Plot Form for Stability Analysis. >



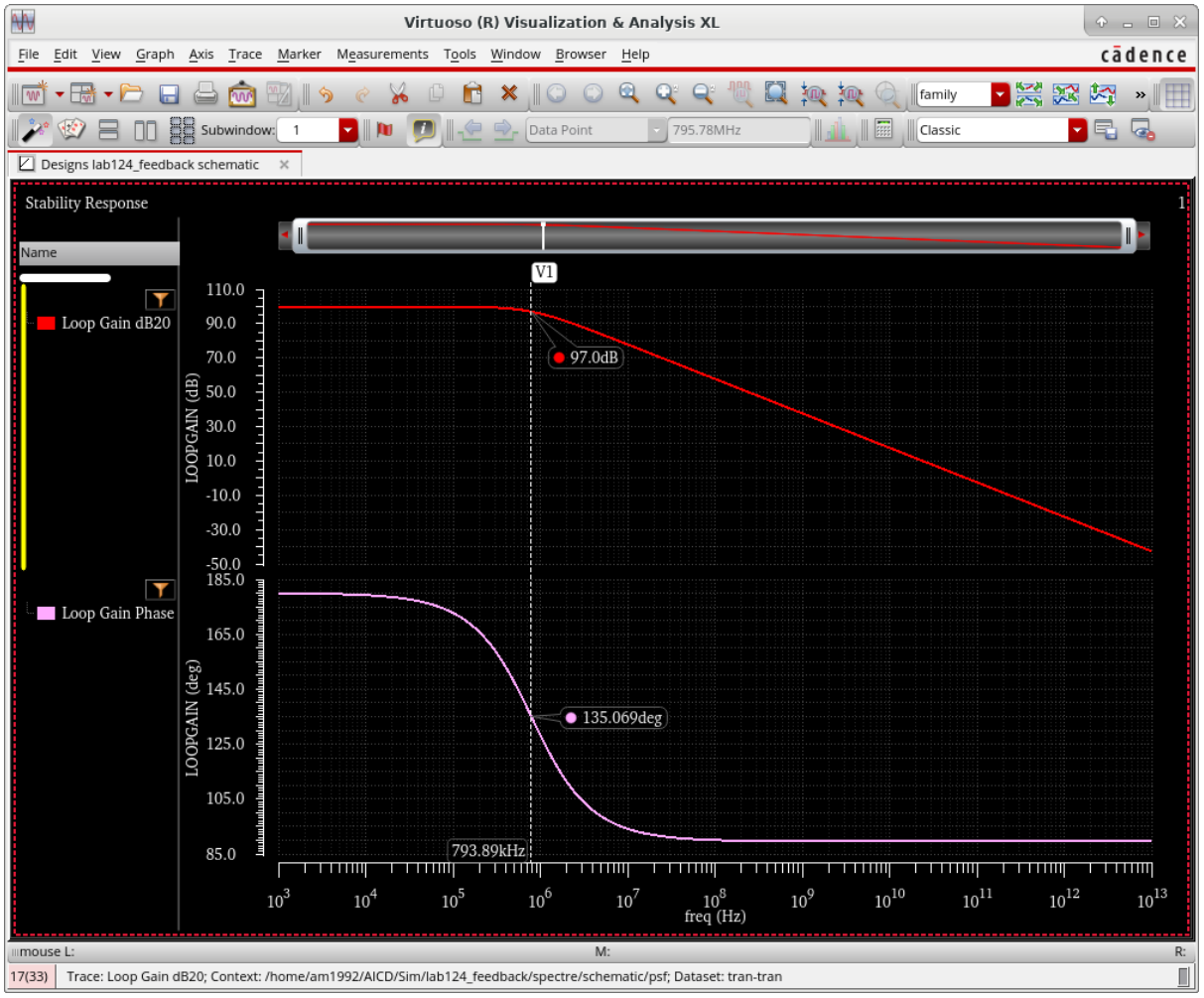
< Figure 98. Loop Gain Magnitude and Phase Plot of the Simple Feedback System. (C0=C1=0). Note that from the frequency response there are no poles or zeros. >

(4.5.8) Note: Referring to Figure 17. Since both the poles in the forward and feedback paths of the feedback system have been removed (C0=C1= 0), there is no frequency dependence of the loop gain magnitude and loop gain phase. The feedback system is unconditionally stable, since the feedback signal is not subject to any phase shift and therefore can never make the 180° transition from stable negative feedback to positive feedback.

(4.5.9) Return the pole to the forward path of the feedback system. Set the forward path capacitor back to 2pF (Capacitance = 2p).

(4.5.10) Check and save your schematic using the **Check and Save** command.

(4.5.11) Re-simulate the stability analysis. Again, use the Direct Plot form to plot the loop gain (dB20) vs. frequency and loop gain phase (degrees) vs. frequency. See Figure 18.



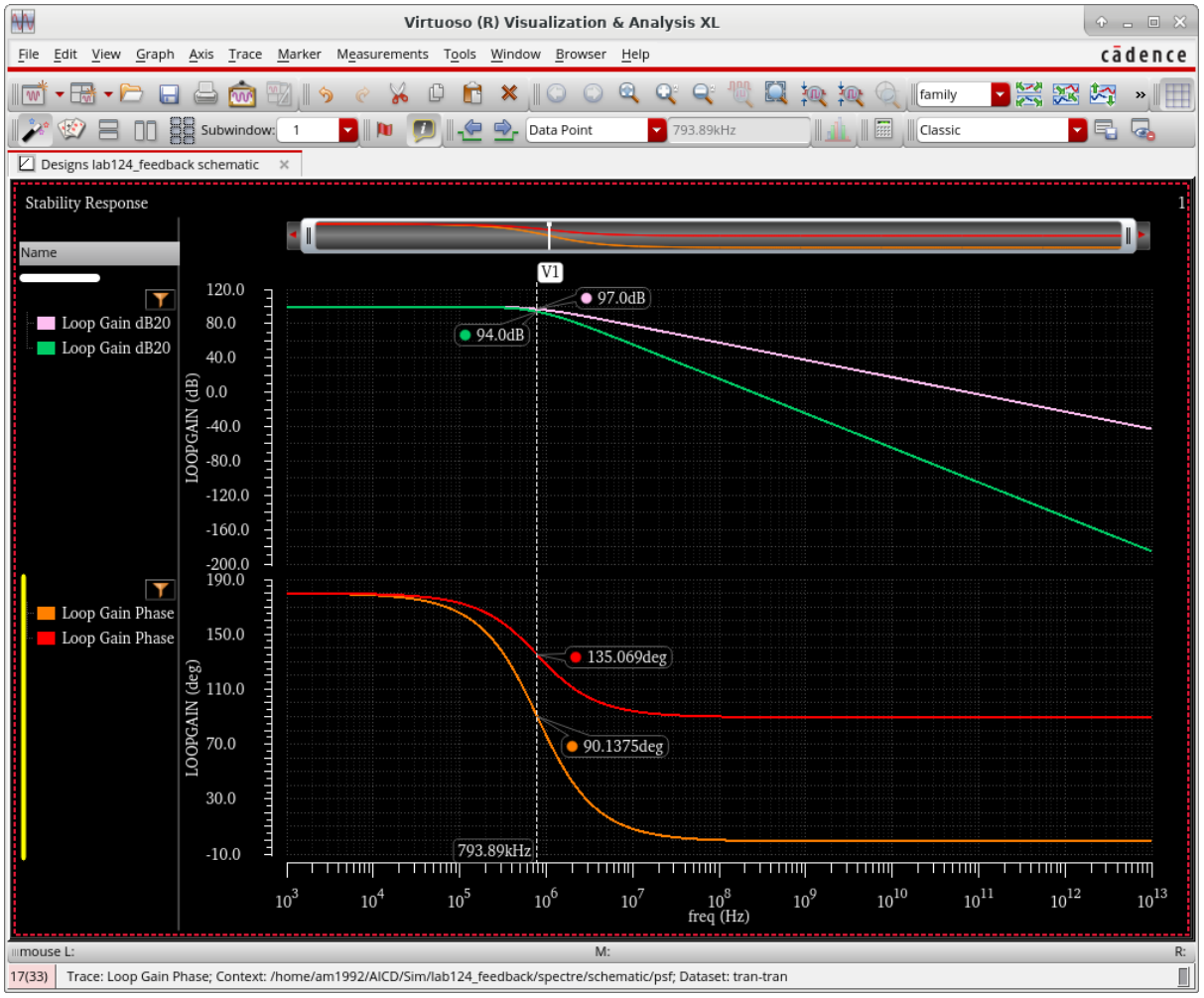
< Figure 99. Loop Gain Magnitude and Phase Plot of the Simple Feedback System. With C0=2pF and C1=0. Note that there is only one pole in the forward path ~794KHz. >

(4.5.12) Note: Referring to Figure 18. With the pole in the forward path of the feedback system (C0 = 2pF, C1 = 0), there is a frequency dependence of the loop gain magnitude and loop gain phase. On the magnitude plot the pole introduces a roll-off of 20dB per decade starting at the corner-frequency given by $\omega_{-3dB} = 1/RC$ where R=100K, C=2pF. (Giving $f_{-3dB} \approx 795.78KHz$). Since a pole can introduce a phase shift of up to 90° in the feedback signal, again, the feedback signal cannot make the 180° transition from stable negative feedback to unstable positive feedback. The feedback system is still unconditionally stable.

(4.5.13) Now, return the pole to the feedback path of the feedback system. Set the feedback path capacitor back to 2pF (Capacitance = 2p).

(4.5.14) Check and save your schematic using the **Check and Save** command.

(4.5.15) Re-simulate the stability analysis. Again, use the Direct Plot form to plot the loop gain (dB20) vs. frequency and loop gain phase (degrees) vs. frequency. See Figure 19.

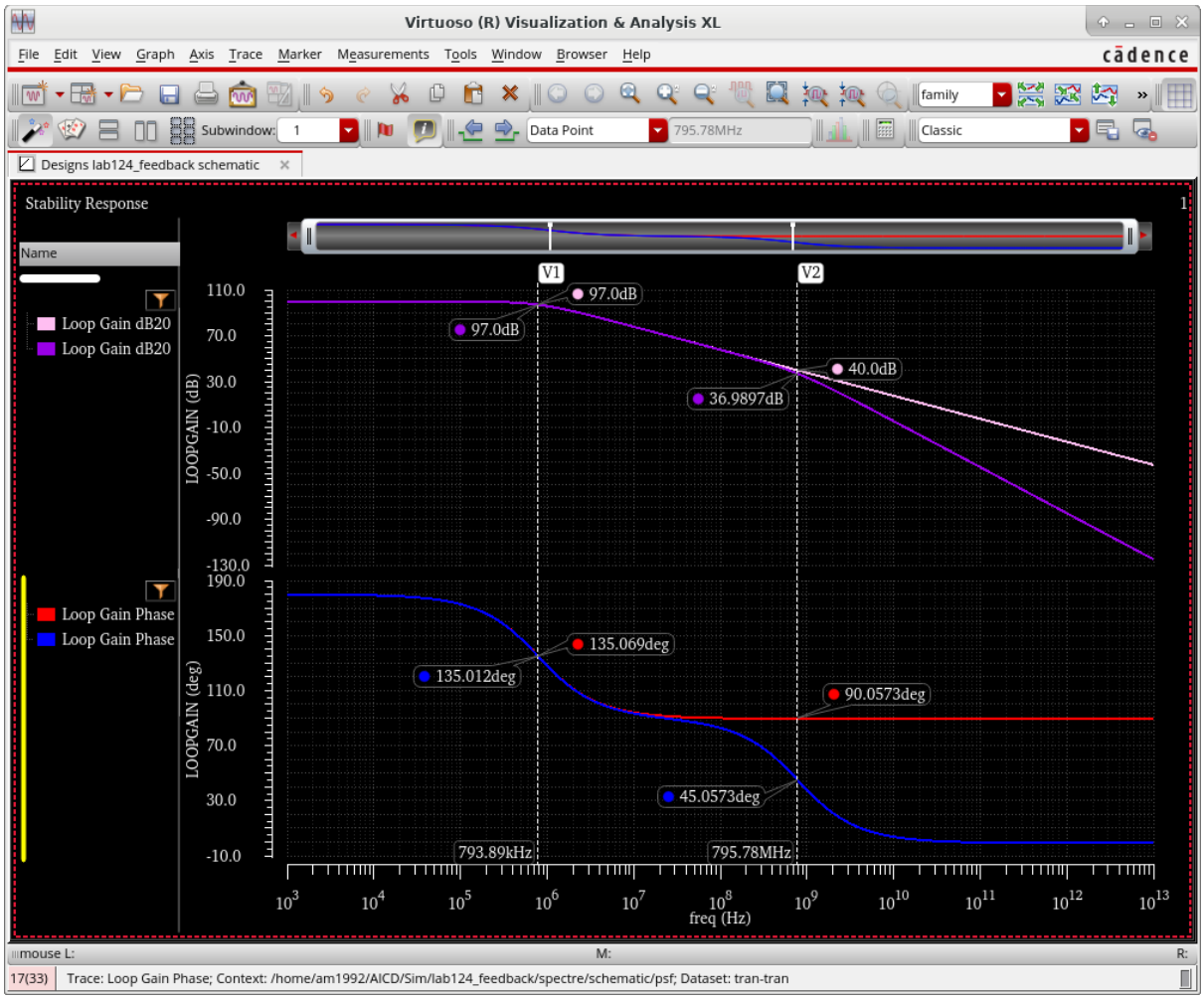


< Figure 100. Loop Gain Magnitude and Phase Plot of the Simple Feedback System. With C0=2pF and C1=2pF. Note the there are two poles: one in the forward path and one in feedback path. Both poles are located at the same frequency ~794KHz, so neither is dominant. >

(4.5.16) Note: Referring to Figure 19. With both the forward path and feedback path poles present in the feedback system (C0 = 2pF, C1 = 2pF). As we would expect, there is a frequency dependence of the loop gain magnitude and loop gain phase.

On the magnitude plot, each pole introduces a roll-off of 20dB per decade and since there are two poles closely located, we see a 40 dB roll-off overall – both poles give rise to the same the corner-frequency, $\omega_{-3dB} = 1/RC$ where R=100K, C=2pF. (Giving $f_{-3dB} \approx 795.78KHz$). Each pole also introduces a phase shift of up to 90° in the feedback signal. Since there are two poles, we now see a (close to) 180° transition in the feedback signal. The feedback system is therefore on the very edge of instability. This system is technically referred to as **marginally stable**, since the feedback never quite becomes positive. Bear in mind, however, that this is a very simple ideal system; for all practical purposes this system would be considered **unstable** (or at least **insufficiently stable**), especially when you consider that in a real system – subject to parasitics, PVT variation, parameter drift over time, etc – a marginally stable system can very easily be tipped into instability.

(4.5.17) As we noted above, both forward path and feedback path poles are closely located. To make the effect of each pole more obvious, we will make the feedback path pole non-dominant.



< Figure 101. Loop Gain Magnitude and Phase Plot of the Simple Feedback System. With C0=2pF and C1=2fF. Note the there are two poles: one in the forward path and one in feedback path. The forward path pole is still located at ~794Khz. The feedback path pole is located at ~796MHz. The forward path pole is dominant. >

(4.5.18) Set the feedback path capacitor to 2fF (Capacitance = 2f). Keep the resistor at 100KΩ.

(4.5.19) Check and save your schematic using the **Check and Save** command.

(4.5.20) Re-simulate the stability analysis. Again, use the Direct Plot form to plot the loop gain (dB20) vs. frequency and loop gain phase (degrees) vs. frequency. See Figure 20.

(4.5.21) Note: Referring to Figure 20. Both the forward path and feedback path poles are present in the feedback system (C0 = 2pF, C1 = 2fF), but one dominates. On the magnitude plot, each pole introduces a roll-off of 20dB per decade, but this time we see the two break frequencies more clearly. The pole in the forward path causes a roll-off of 20dB per decade and dominates the frequency response since it occurs at a much lower frequency, the corner-frequency set by $\omega_{-3dB} = 1/RC$ where R=100K, C=2pF. Giving $f_{1\text{-}3dB} \approx 795.78KHz$. The second pole occurs at a much higher frequency, causing the roll-off to increase to 40 dB per decade, the second break frequency occurring at $\omega = 1/RC$ where R=100K, C=2fF. Giving $f_{2\text{-}3dB} \approx 795.78MHz$ Changing the location of the second pole (in the frequency domain) does not change the stability of the feedback system. Again, each pole introduces a phase shift of 90° in the feedback signal. We still see (close to) a 180° transition in the feedback signal, with the feedback system still on the edge of instability. This system would still be considered unstable.

(4.5.22) We will continue our simulations of system stability in part two of this lab using a different design/schematic. We will discuss the general approach to compensating unstable systems and we will introduce the stability metrics of phase margin and gain margin.

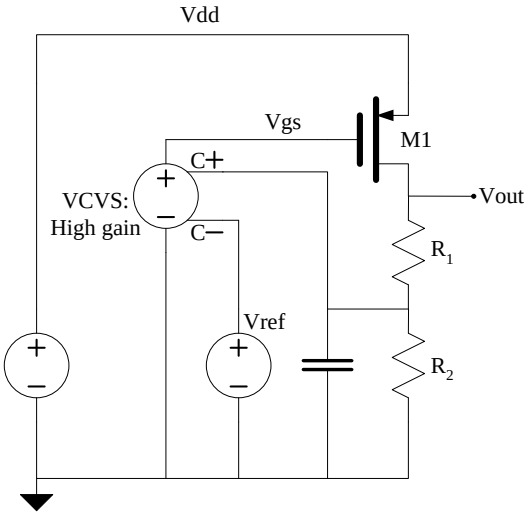
5. Circuit Simulation Exercises: Part 2

With guidance, you will:

- Investigate the stability of a voltage reference circuit with modelled op-amp drive.

5.1 A Controllable Voltage Reference Circuit

(5.1.1) Create the schematic for the IC voltage reference circuit in Figure 21. A summary of the circuit's operation is given in section 4.6 of the background material.

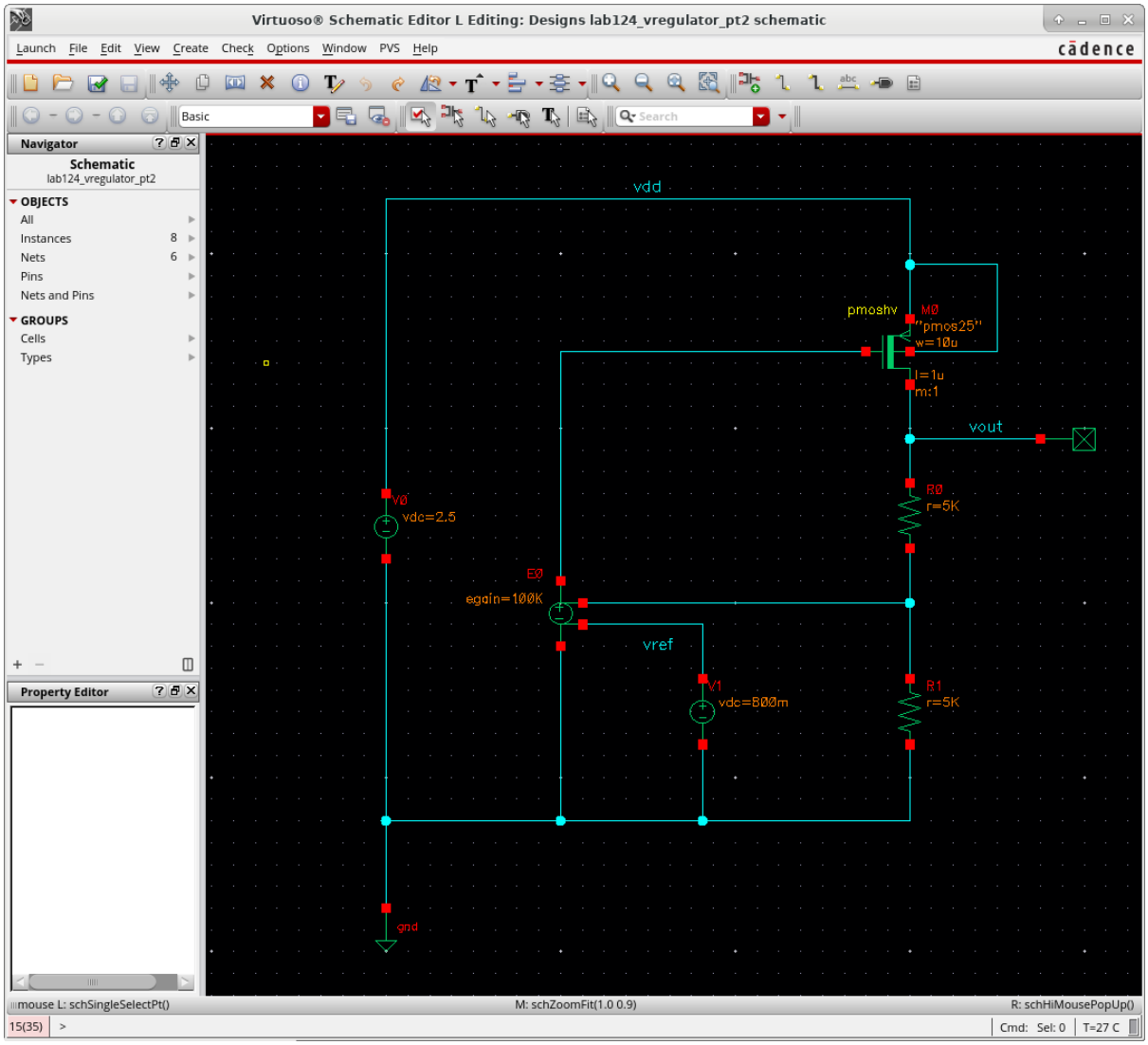


< Figure 102. IC Controllable Voltage Reference Circuit
Reproduced from Figure 30, section 4.6 of the background material. >

(5.1.2) From the Library Manager window, select the 'Designs' library and choose **Menu: Files à New à Cell View** to invoke the New File form. Give the new schematic a meaningful name such as lab124_vregulator_pt2.

(5.1.3) In the schematic editor window, construct the circuit given in Figure 21 (22). From the analogLib library instantiate and place two DC voltage sources (**vdc**), two resistors (**res**), a voltage-controlled voltage source (**vcvs**), and a ground reference (**gnd**). Also instantiate one high-voltage PMOS FET (**pmoshv**) from the GPDK180 library. Set the DC supply voltage to 2.5V (**DC Voltage** = *2.5*), set the reference voltage to 0.8V (**DC Voltage** = *0.8*), set the two resistors to 5KΩ each (**Resistance** = 5K), and dimension the PMOS device 10um/1um (**width** = *10u*, **length** = 1u). Leave all other attributes unchanged.

(5.1.4) Wire the components as in Figures 21 and 22 using the **Wire (narrow)** command. **Remember to connect the body/bulk terminal of the PMOS device to the most positive net in the circuit (VDD=2.5V)**. Using the **Create Wire Name** command, name the DC supply net *vdd*, the PMOS gate net *vgs*, and the reference input net *vref*.



< Figure 103. Controllable Voltage Reference Circuit. >

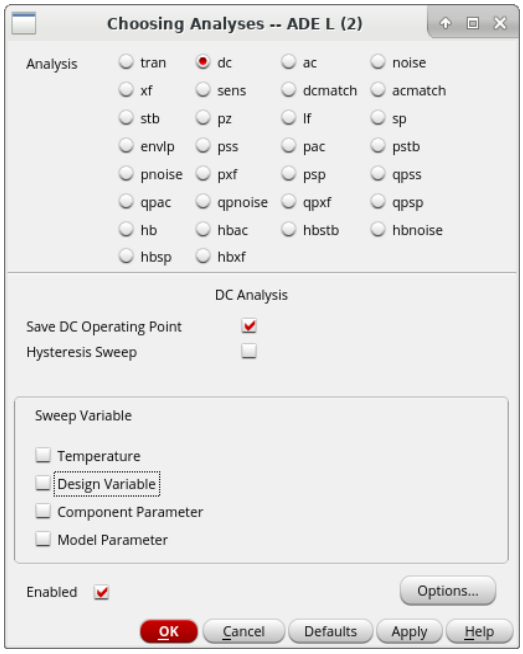
(5.1.5) Check and save your schematic using the **Check and Save** command. Remember to use the output area of the CIW to diagnose any problems with your schematic and/or simulations if there are any warning or error messages when checking, saving, or simulating your work.

5.2 DC Operating Point Analysis of the Volage Reference Circuit

(5.2.1) You should now have the completed schematic of the voltage reference circuit as shown in Figure 22. It should check and save without any errors or warnings.

(5.2.2) We will simulate the voltage reference using a DC Analysis. From the Schematic Editor window, launch the Analogue Design Environment (ADE) from **Menu: Launch à ADE L**.

(5.2.3) From the ADE window, go to **Menu: Analyses à Choose…** or use the **Choose Analyses** button on the right-side of the ADE window (top button). The 'Choosing Analyses – ADE L' form will open. On the form, select the '**dc**' option button. Check the 'Save DC Operating Point' box. OK the form.

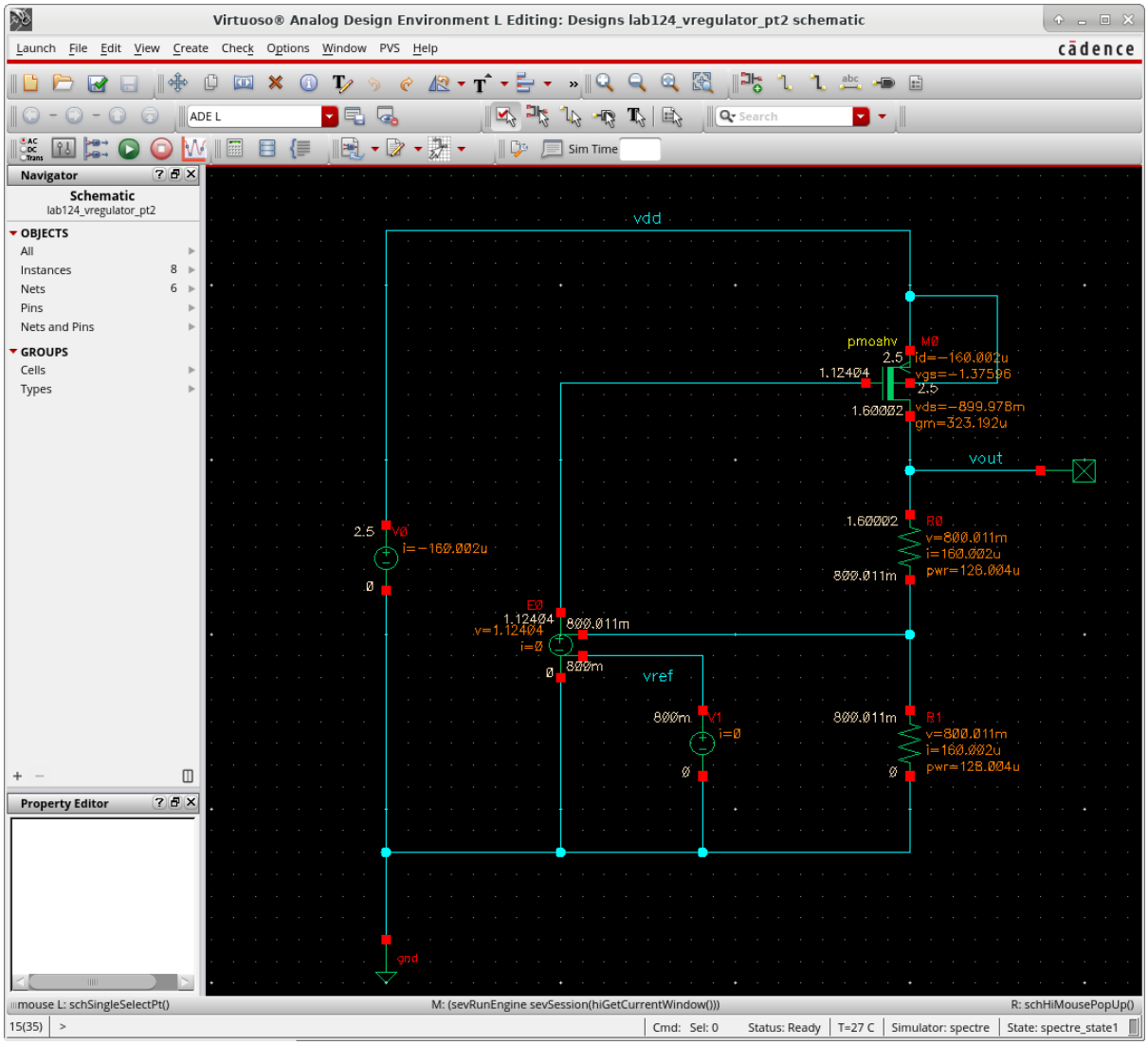


< Figure 104. The Choosing Analyses Form. Only DC Operating Point selected. >

(5.2.4) At this stage we are only simulating for the DC operating point. Simulating the DC operating point does not produce an output waveform, so we do not need to select any outputs to be plotted.

(5.2.5) In the Analog Design Environment window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error. The text output window should appear but there will be no waveform window.

(5.2.6) From the ADE window choose **Menu: Results à Annotate à DC node voltages…** If you now look in the schematic window, you should see the node voltages annotated directly on the schematic. Again, from the ADE window, choose **Menu: Results à Annotate à DC operating points…** You should now see (along with the node voltages) additional DC operating point information annotated on the schematic including the DC bias currents. See Figure 24.



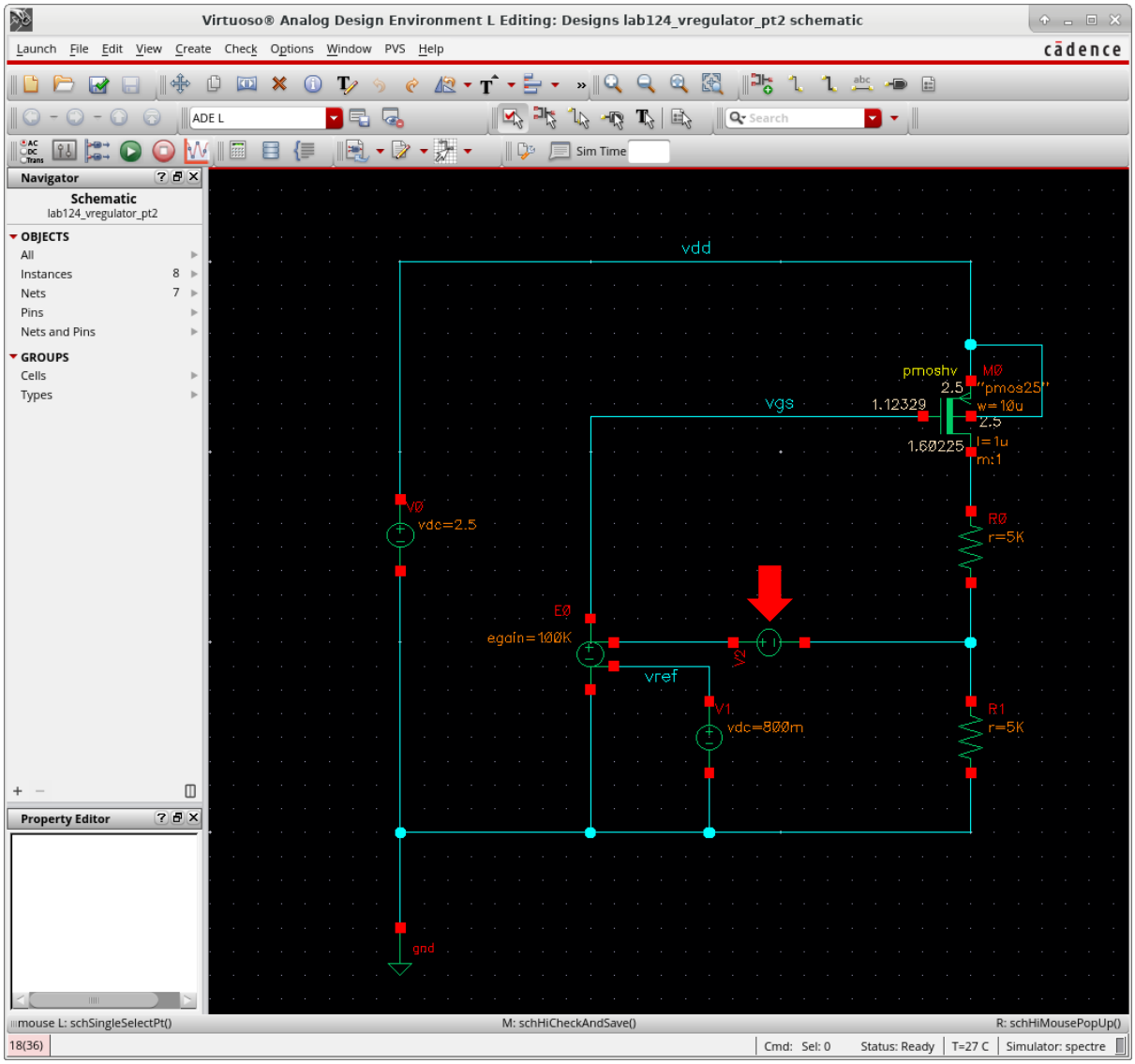
< Figure 105. Controllable Voltage Reference Circuit with Node Voltage and DC Operating Point Annotations. >

(5.2.7) **Discussion**. Refer to section 4.6 of the background material for a discussion on circuit operation. The important point to note is that the high gain of the VCVS regulates the output voltage(*vout*) to be twice that of the voltage reference (*vref*). The factor of two occurs due to the even division of the feedback signal across the potential divider. The feedback results drives the same voltage to be on both VCVS inputs, this requires an output voltage across the resistors equal to 2*vref so that vcvs(+) equals vcvs(-). The high gain of the VCVS is used to overcome any non-ideal behaviour of the PMOS – or any other disturbance. The higher the gain the more closely the output will track the voltage reference (by a factor of two.) Note that the PMOS device provides an inversion in the loop – so the input to the VCVS is the negative terminal and the feedback signal goes to the positive terminal.

5.3 Stability Analysis of the Volage Reference Circuit

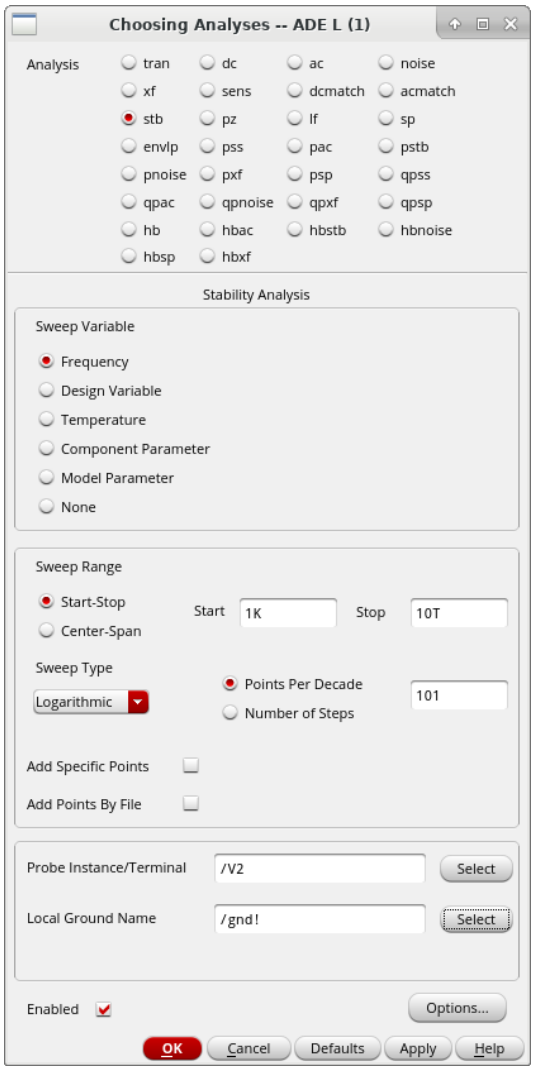
(5.3.1) Here, we will explore the stability of the voltage reference circuit.

(5.3.2) Insert a null **vdc** voltage source in the feedback path just before positive terminal of the VCVS. The AC and DC magnitude values of this component should be set to zero or left empty. See Figure 25 (highlighted by the red arrow). This vdc does not in any way change the functionality of the circuit or its output response. It is used only as a loop identifier/probe by the stability analysis tool.



< Figure 106. Annotated Controllable Voltage Reference Circuit with null VDC component in the feedback path for stability analysis. >

(5.3.3) We want to determine the stability of the feedback system by examining how the loop gain and phase of the circuit change with frequency.



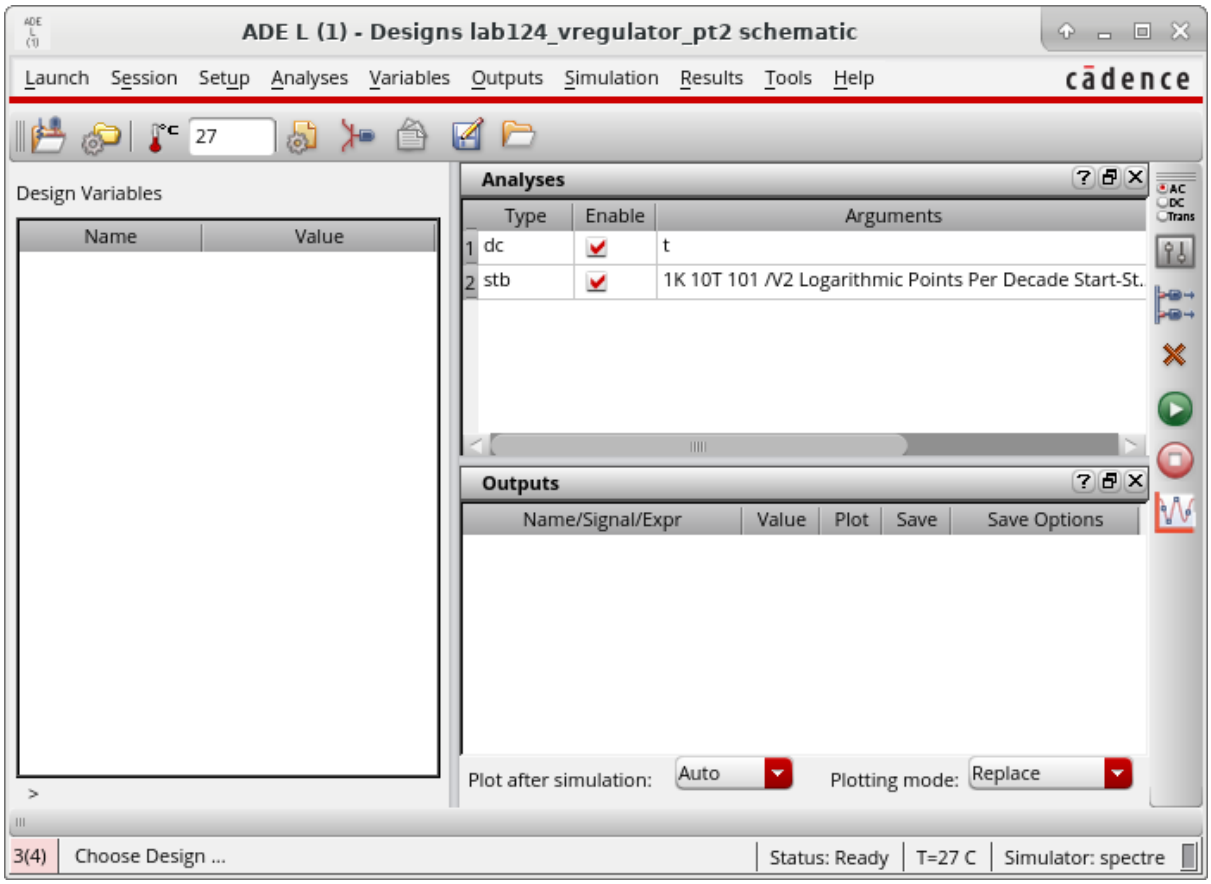
< Figure 107. The Choosing Analyses Form for a Stability Analysis. >

(5.3.4) From the ADE window, setup a stability analysis (**Menu: Analyses à Choose**). In this simulation we will again sweep the frequency from 1KHz to 10THz using a logarithmic sweep of the frequency – with 101 plot points per decade of frequency. At this stage, we will not select any outputs for plotting (we will use the Direct Plot Form later).

In the choosing analyses form, select the '**stb**' option button. Select the 'Frequency' option as the sweep variable. Choose the 'Start-Stop' option button with Start = 1K and Stop = 10T for the sweep range. Select 'Logarithmic' from the drop-down menu as the sweep type. Choose the 'Points Per Decade' option button with 101 plot points per decade.

At the bottom of the form, there are two fields: 'Probe Instance/Terminal' and 'Local Ground Name'. The probe instance/terminal is used to identify the feedback loop – we need to point this to the null vdc component we inserted into the feedback path. Press the select button next to the 'Probe Instance/Terminal' form field. The schematic should come into focus. Click on the null vdc component. The name of the null vdc part (/V2) should now appear in the form field. We also need to do the same with the local ground name. Press the select button next to the 'Local Ground Name' form field. The schematic should come into focus. Click on any net connected to a ground (gnd) reference. The name of the ground net (/gnd!) should now appear in the form field.

Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. OK the form. The **stb** analysis should appear in the 'Analyses' list of the ADE window next to the **dc** analysis already set up. See Figure 27.

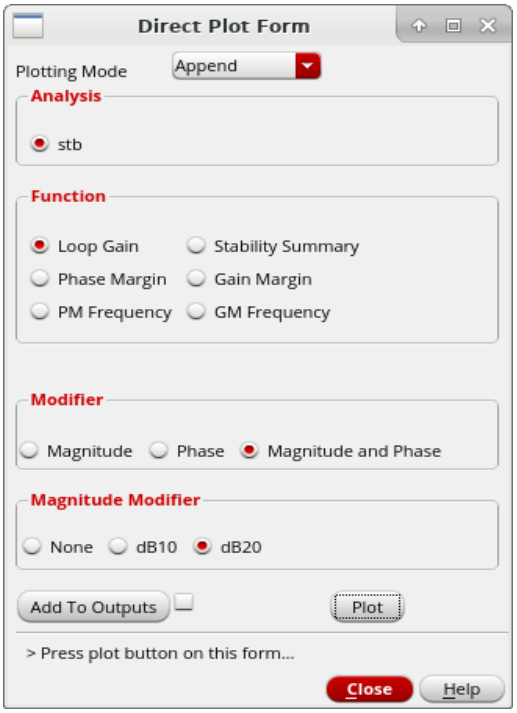


< Figure 108. The ADE with the DC OP Analysis and STB Analyses set up. >

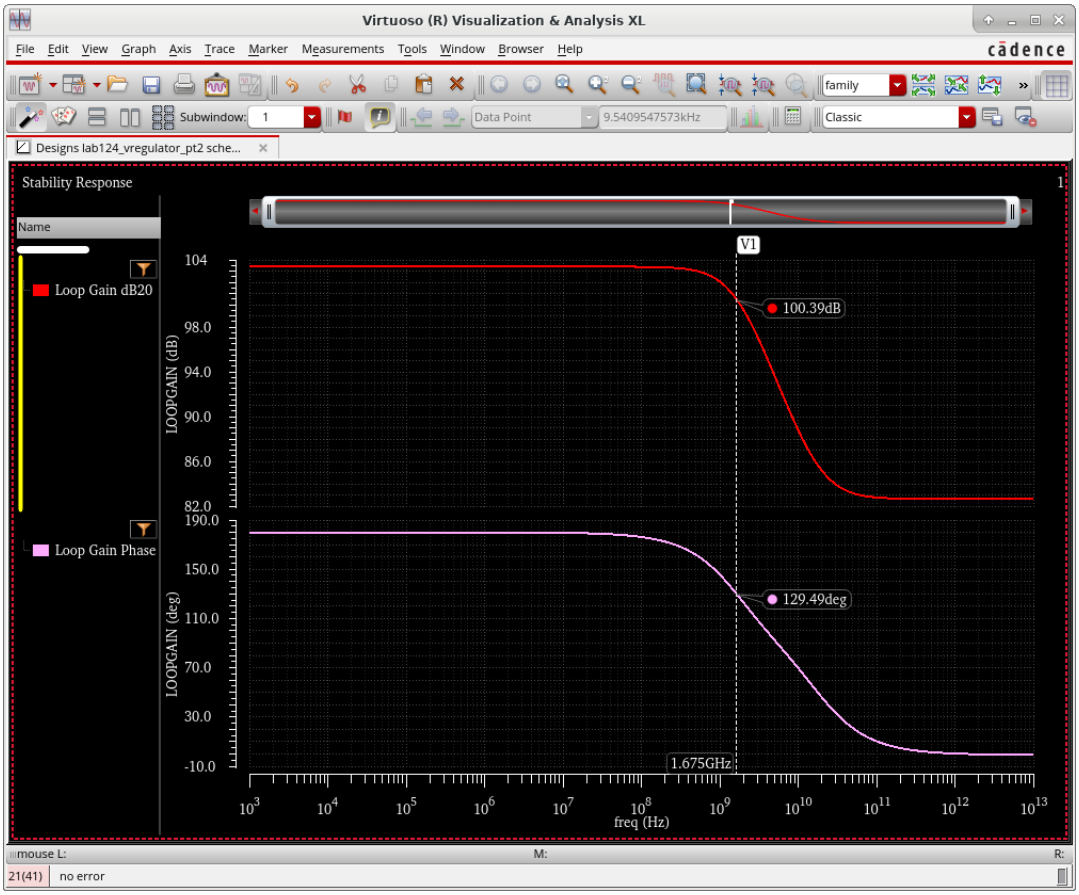
(5.3.5) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error. Only the text output window should open automatically since we didn't select an output for plotting.

(5.3.6) From the ADE window, open the Direct Plot form (**Menu: Results → Direct Plot → Main Form**). Only the **stb** analysis will show up on the Direct Plot Form since the **dc** operating point analysis doesn't have any plotting options.

Make sure the **stb** option button is selected in the Analysis section of the Direct Plot Form. Select the **Loop Gain** option button from the function section. Select the **Magnitude and Phase** option in the modifier section of the direct plot form. In the magnitude modifier section, select the **dB20** option. Click the 'Plot' button near the bottom right corner of the form. You should now have both a plot of loop gain in dB20 vs. frequency and a plot of loop gain phase (degrees) vs. frequency plotted in the waveform window. See Figures 28 and 29.



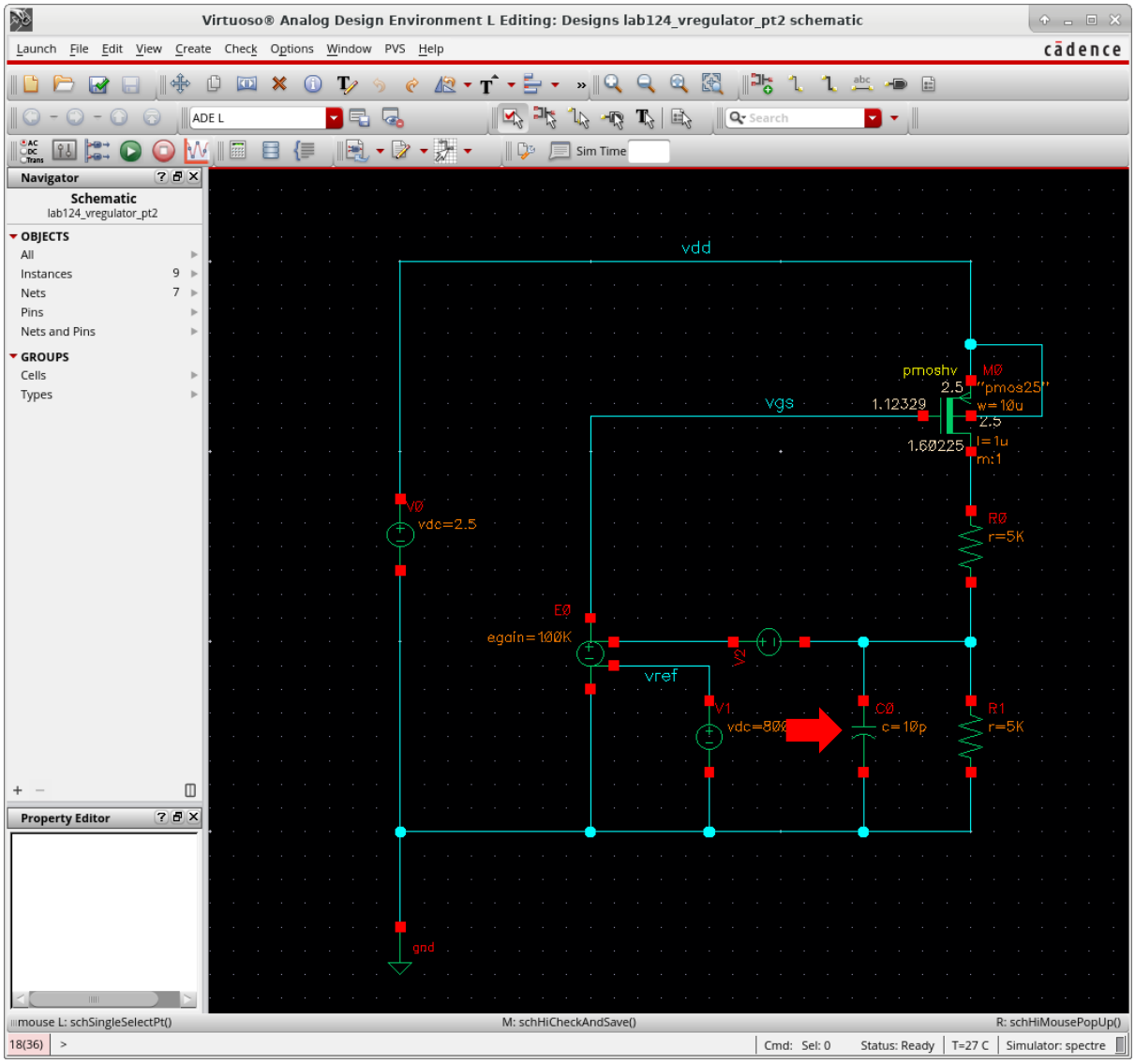
< Figure 109. The Direct Plot Form for the Stability Analysis>



< Figure 110. Loop Gain Magnitude and Phase Plot of the Controllable Voltage Reference Circuit. The DC gain is ~103.39dB. With Giving f_{-3dB}≈1.675GHz >

(5.3.7) Split the traces for clarity. Right-click on the empty area of the graph window and select **Split Current Strip à trace** from the context menu.

(5.3.8) **Disscussion**. We see from the stability analysis that the voltage reference is unstable. The feedback signal undergoes an almost 180° phase shift, with the gain remaining very high throughout. Clearly, at the higher frequencies the feedback becomes (almost) positive. Techincally, the system is marginally stable since the phase shift doesn't quite reach 180° – but for all practical perposes it would be considered unstable. As we have not introduced any recative components into the circuit, this instability occurs entirely due to the parasitic recatances modelled in the PMOS device – of which there is more than one.

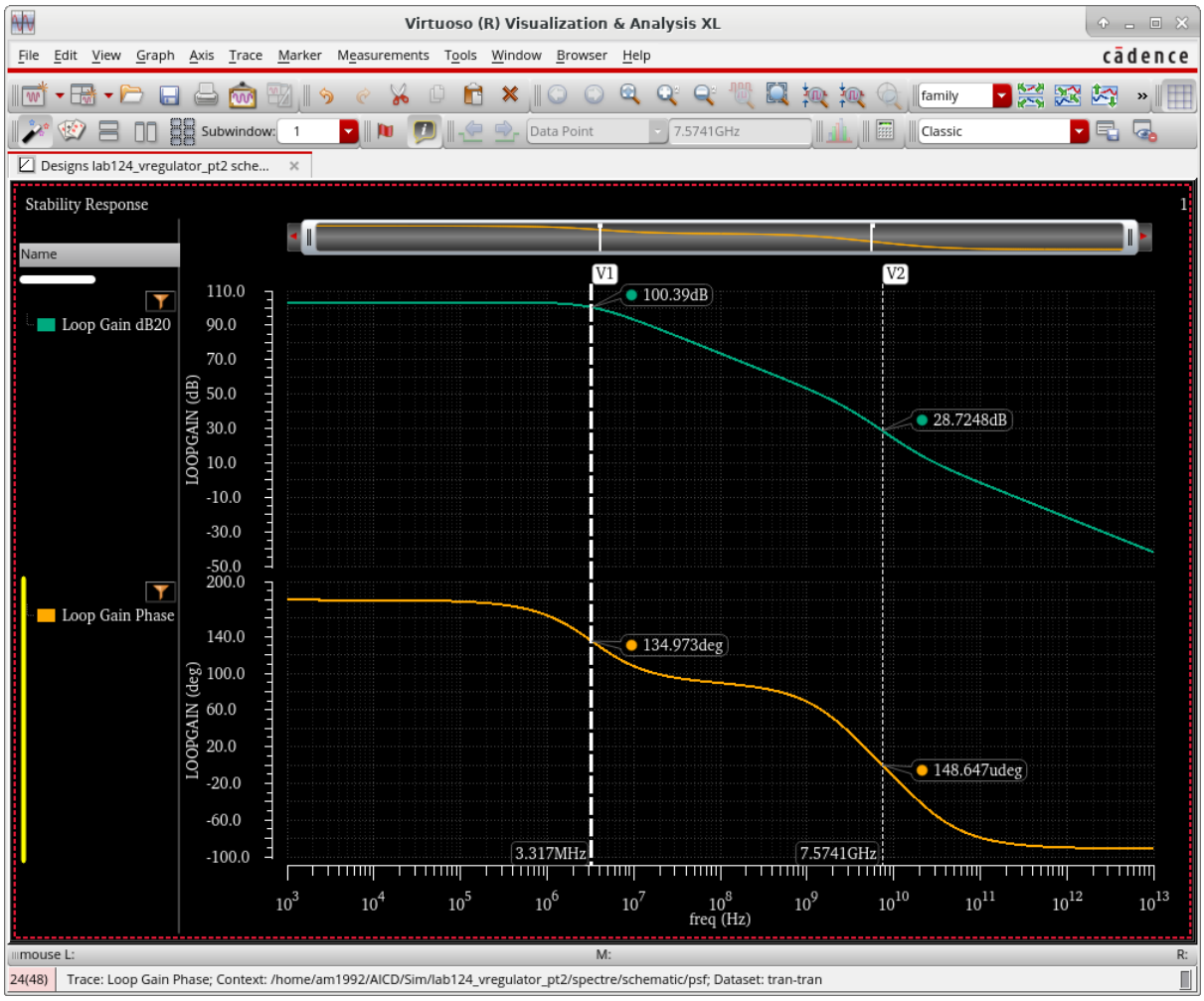


< Figure 111. Annotated controllable voltage reference circuit with null VDC and 10pF compensation capacitor in the feedback path to improve stability. >

(5.3.9) Recall, that to ensure stability we want the loop gain to fall below 0dB before the phase shift reaches 180°. We can either (1) lower the gain of the VCVS feedback amplifier (though, this would adversely affect the DC tracking accuracy of the reference), or (2) insert a dominant pole into the feedback path to bring down the gain as the frequency increases. The capacitor will form a pole with the two resistors of the potential divider and lead to a roll-off in the gain. If the gain can be brought down to below 0dB before the loop gain phase shifts 180° we would have stabilised the system.

(5.3.10) To improve the stability we will first insert a capacitor to ground in the feedback path. From the analogLib instantiate a capacitor (**cap**). Set the capacitance to 10pF (**Capacitance** = 10p). Place it into the feedback path as in Figure 30.

(5.3.11) Re-simulate the stability analysis. Again, use the Direct Plot form to plot the loop gain (dB20) vs. frequency and loop gain phase (degrees) vs. frequency. See Figure 31.

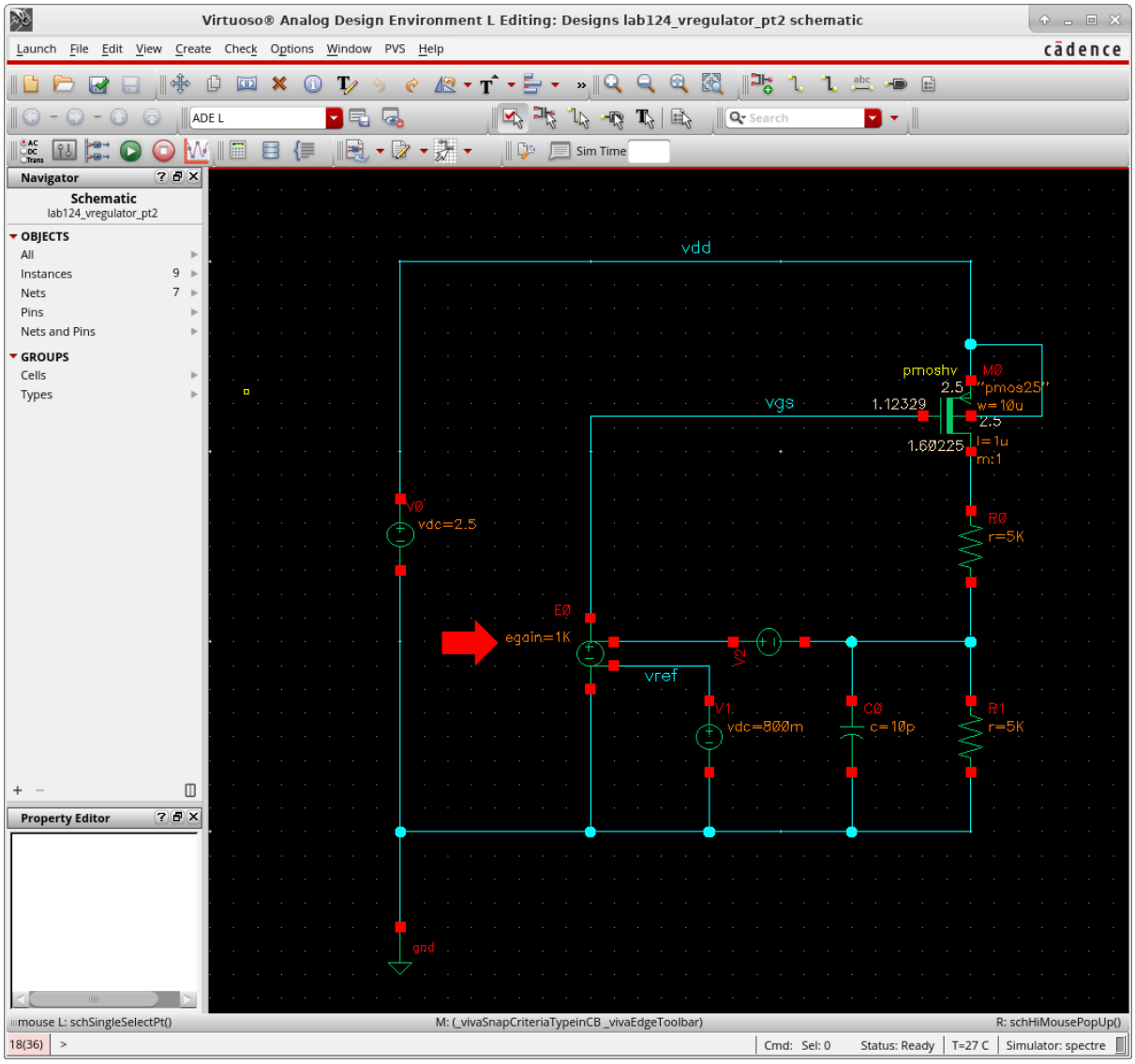


< Figure 112. < Figure 113. Loop Gain Magnitude and Phase Plot of the Controllable Voltage Reference Circuit with 10pF Compensation Capacitor. The DC gain is ~103.39dB. With Giving f_{-3dB}≈3.317MHz. Still unstable. >

(5.3.12) **Discussion**. Even with the 10pF capacitor in the feedback path the voltage reference is still unstable. We have a loop gain of over 28dB as the loop gain phase-shift reaches 180º. This is an improvement compared to the no-capacitator version (82dB, see Figure 29). Since capacitors take up a lot of space, increasing the capacitance in the feedback path isn't usually the preferred option (5pF to 10pF is an upper limit of what is acceptable). Instead, we will lower the gain of the VCVS. This is a trade-off in tracking accuracy vs. stability.

(5.3.10) Modify the VCVS voltage gain from 100K to 1K. *You can use **Menu: Edit → Properties → Objects**, the '**Edit Properties**' icon-button from the toolbar, or the keyboard shortcut '**Q**'. You can then modify the required attribute(s).*

(5.3.11) Re-simulate the stability analysis. Again, use the Direct Plot form to plot the loop gain (dB20) vs. frequency and loop gain phase (degrees) vs. frequency. See Figure 34.

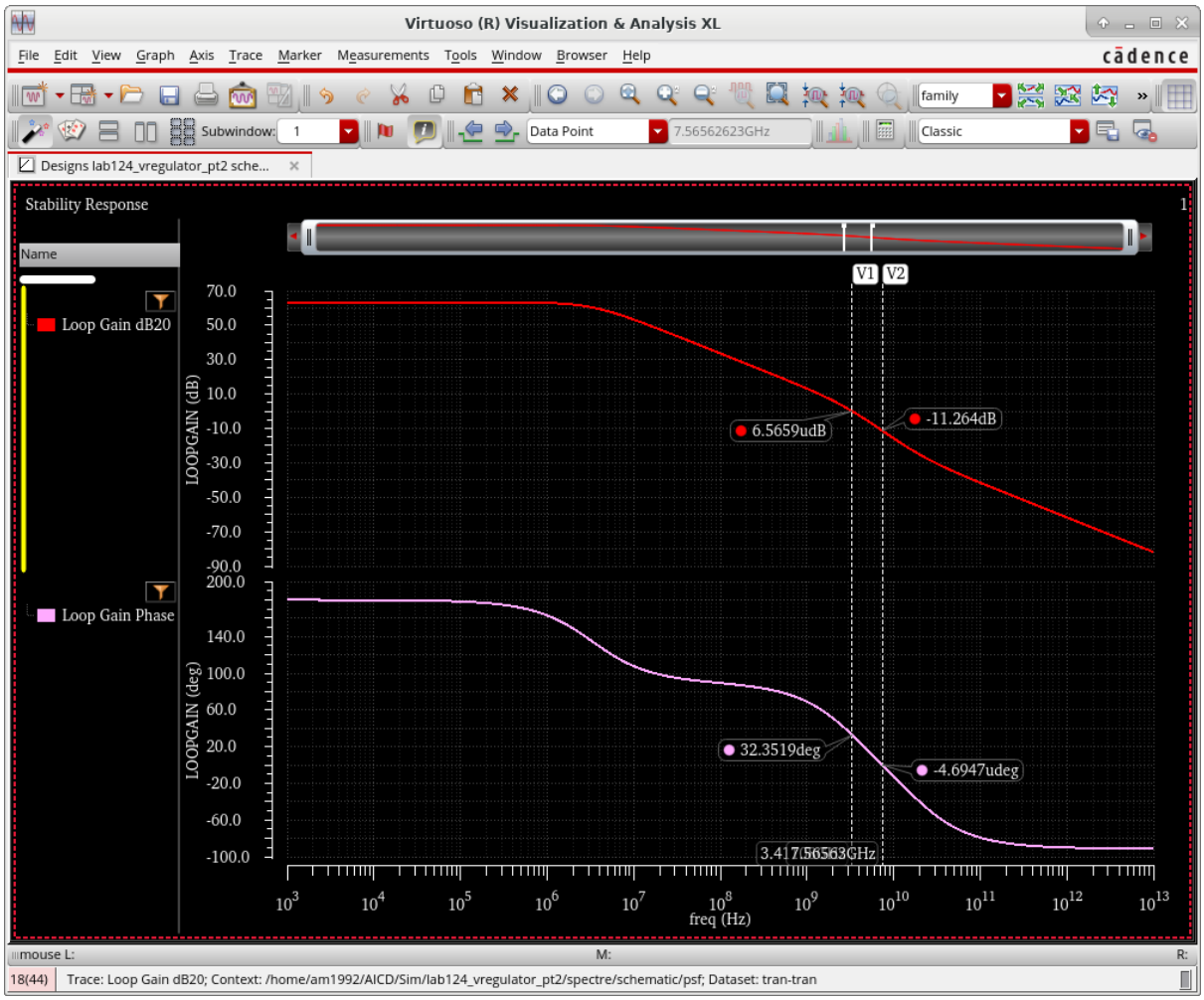


< Figure 114. Annotated controllable voltage reference circuit with null VDC. Includes 10pF compensation capacitor in the feedback path and a lowered VCVS gain (from 100K to 1K) to improve stability. >

(5.3.12) **Discussion**. With the lower VCVS gain (1K) and the 10pF capacitor in the feedback path, the voltage reference is now stable, at a cost of a slightly lower tracking accuracy. Note the new DC operating point of the voltage reference – previously 1.6002V with the VCVS=100K, now 1.602V with the VCVS=1K. This would be considered, in most cases, an acceptable trade-off (but this would really depend on the required specification of the reference.)

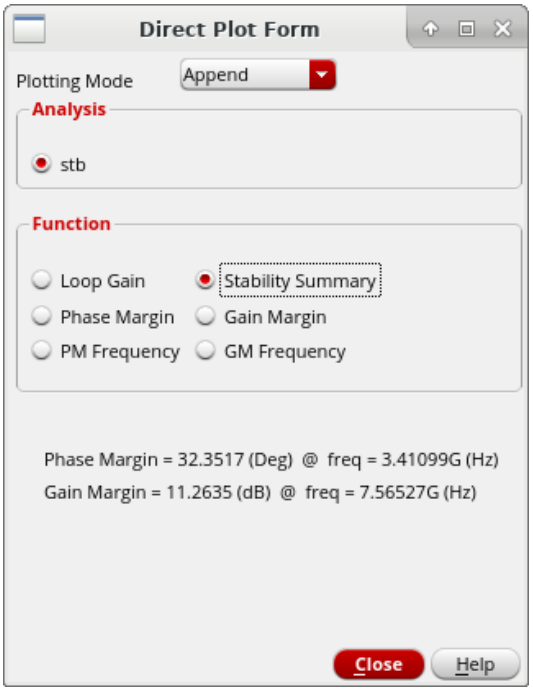
Note that two vertical markers have been placed on the magnitude and phase plots of the loop gain. When the gain falls to 0dB (6.57udB) note that the phase has shifted from 180° to 32.3519°. 32.3519° is therefore the phase margin. When the phase has shifted fully to 180° the gain has fallen to -11.264dB. 11.264dB is therefore the gain margin.

Whilst it is now stable, the phase margin is only 32°. As a rule of thumb, we would aim for a phase margin of 60° to 70° – this provides a good performance trade-off between system responsiveness (in the time domain) and stability in the face of PVT, parameter drift, and so on. (Appendix C in [2] offers a good development of the relationship between phase margin and time domain response).



< Figure 115. Loop Gain Magnitude and Phase Plot of the Controllable Voltage Reference Circuit with 10pF Compensation Capacitor and Lowered VCVS Gain (1K). Stable, with a Phase margin = 32° and a Gain Margin of 11dB. >

(5.3.13) Finally, on the direct plot form select **Stability Summary** from the function section. Note that the gain and phase margin (at their recpective frequncies) are displayed.



< Figure 116. The Direct Plot Form Displaying the Stability Summary. >

End of Lab!